

December 20, 2013

US Interest Rate Strategist Hard Pitch, Soft Ball

The transition toward a less credible rate guidance framework watered down the Fed's stronger rate guidance. Less effective rate guidance and stronger growth data should allow intermediate maturity Treasuries to underperform into the new year.

US Governments

The rate path priced by the market will require a higher level of term premium – a dynamic that is likely to manifest most strongly in the 5- to 7-year sector. We suggest going short the 7-year point on the 3s/7s/10s butterfly (50:50 weighted), or going short the 5-year point on the 2s/5s/30s butterfly (50:50 weighted).

US Treasury Inflation-Protected Securities

We turn neutral on TIPS breakevens. Federal Reserve officials signaled they are unlikely to change the rate guidance framework in the near term. Without a more accommodative change, we see little reason why TIPS breakevens should rally.

US Short Duration Strategy

We expect repo rates to trade in the 8-15bp range in the first quarter of 2014. The Fed announced that it will taper asset purchases, but provided more accommodation by strengthening forward guidance and lowering the projected target fed funds rate. We expect the Fed to extend the reverse repo facility next year.

US Interest Rate Derivatives

High realized vol pushed the upper-left of the vol surface higher. We examine how static guidance may affect vol going forward. A 4s/7s/10s payor butterfly hedges continued underperformance in belly rates, delivers optimal risk-adjusted rolldown and carry, and offers better value than a simple long payor position.

US Agency Mortgage-Backed Securities

The Fed is still expected to buy more than our 2014 net issuance projection. MBS performance is less likely to be driven by pure OAS widening alone next year. We are most concerned about bear-flattening of the 5-10y portion of the curve, an increase in volatility, and swap spread widening. We recommend entering a FN 4 Fly.

Please note that our next publication will be on January 10, 2014. Happy Holidays!

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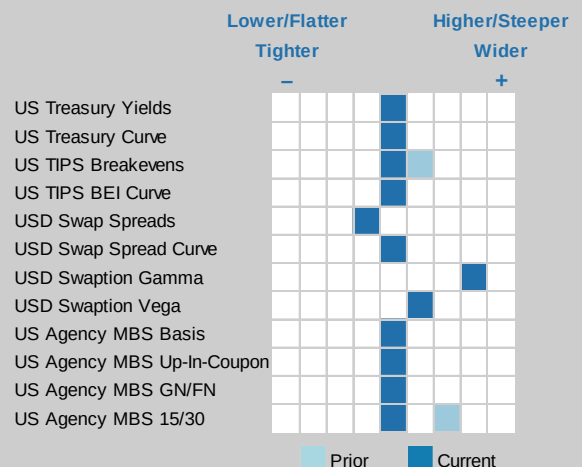
Ayan Nasir

Regular Features

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Global Strategists

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US Interest Rate Strategist

Our Views

Our Views	Suggested Strategies
US Treasuries	
- Given the uncertainty around what determines the Fed's collective view once the unemployment rate threshold is crossed, the market is left to judge today's rate guidance on its credibility – i.e., the probability the Fed will not renege on today's rate guidance tomorrow. - The rate path priced by the market will require a higher level of term premium – a dynamic that is likely to manifest most strongly in the 5- to 7-year sector. We suggest going short the 7-year point on the 3s/7s/10s butterfly (50:50 weighted), or going short the 5-year point on the 2s/5s/30s butterfly (50:50 weighted). - For a less directional play that retains the potential to profit from lower rate guidance credibility, we suggest entering a 3s/5s curve flattener versus a 10s/30s curve flattener (1.0:0.5 DV01 weights of curves).	3s/5s curve flattener versus a 10s/30s curve flattener (1.0:0.5 DV01 weights of curves). - Short the 7-year point on the 3s/7s/10s butterfly (50:50 weighted), - Short the 5-year point on the 2s/5s/30s butterfly (50:50 weighted).
US Treasury Inflation-Protected Securities	
- Ever since September, when the White House announced the nomination of Janet Yellen as the next chairperson of the Federal Reserve, we've been bullish on TIPS breakevens. The basis of our view was that a Yellen-led Fed would be biased to (1) provide more accommodation to bring down unemployment more quickly and (2) tolerate some above-target inflation. - Our bullish view was also based on what we saw as an increasing probability the Fed would change one of the rate guidance thresholds, providing the market with a hard commitment to this strategy. However, Wednesday's FOMC statement signaled that the committee is not ready to change the thresholds. Based on this and still low realized inflation, we see little reason why TIPS breakevens should rally – at least for now.	Neutral
US Interest Rate Derivatives: Swaps	
- In the belly, fund flows, corporate issuance, and a lack of convexity hedging have driven spreads. Technical and regulatory factors continue to weigh on the back end. 10-year swap spreads are historically related to rate expectations and the ratio of Treasuries to investment grade debt. Our model explains over 70% of the variance in spreads, and generates positive returns in backtested trading strategies.* - Back-end spreads steadily tightened over the past few months due to positioning and corporate issuance. We remain in tighteners, but with spreads closer to our estimate for fair value, move to a more tactical than strategic outlook.	10-year spread tighteners
US Interest Rate Derivatives: Vol	
- We are bullish on vega and gamma. Vol is near the cheapest levels to term premiums, due to a muted outlook for realized vol. Weaker forward guidance should support realized vol as markets try to estimate the Fed's reaction function. - We expect vol in the belly of the surface (5-7 year tails) to outperform as those rates appear sensitive to changes in forward guidance.	6m 5s/30s bull steepeners 1m10y 1x2 payors 3m5y5y 1x2 payors 3m 4s/7s/10s payor flies
US Agency Mortgage-Backed Securities	
- We remain neutral on the mortgage basis. Mortgage valuations look reasonable vs. 7-year swaps, but rich vs. 10-year Treasuries. We recommend MBS investors hedge their duration exposure using swaps instead of Treasuries. - We like the belly of the 30-year coupon stack and recommend overweighting 4s across the coupon stack.	- Overweight 4s across the 30-year stack - Long FN 4.0 Fly

*The performance data provided is a hypothetical illustration of mathematical principles, it does not predict or project the performance of an investment or investment strategy. Past performance is no guarantee of future results.

In Case You Missed It

Key Themes	Detailed Coverage
Top 10 Surprises for 2014	December 13, 2013
To wind down the year, we propose 10 events for the liquid US interest rate markets that could catch market participants by surprise. These surprises do not reflect our base case views, but are meant to highlight risks to our views.	- UST 2-Year Yields End 2014 > 2.00% - UST 10-Year Yields End 2014 < 2.00% 5s/10s Curve Breaches All-Time Highs - TIPS Implied Deflation Probabilities Rise 10y20y Real Rates Rise Above 2.5% - Swap Spread Curve Goes Fully Negative 30-Year Skew Turns Normal - Front End Rates Go Negative - Mortgage CC OAS Approaches Zero - Up-in-Coupon and 15s/30s Underperform Download Report
Global Rates Outlook: Great Expectations	December 6, 2013
- Our first theme for 2014 involves the ability of central banks – notably the Federal Reserve – to deliver credible forward guidance on the path for policy rates. Success or failure will help determine the path for interest rates next year. - Our second theme involves a continued rise in interest rates across most major government bond markets. In fact, we only see five markets where yields will end 2014 lower than where we think they will end 2013: Belgium, Spain, Italy, Ireland, and China. - Our third theme involves higher implied interest rate volatility. The increase in rates we forecast suggests a commensurate increase in implied volatility. We also believe the range of possible outcomes is wider than what current markets imply.	- United States - Euro Area - United Kingdom - Japan - Asia ex-Japan - Australia and New Zealand Download Report
Are You Gellin'?	November 15, 2013
We remain neutral on duration but embrace yield curve steepeners. We believe the yield curve can steepen between the belly and the longer end. We like 5s/10s and 5s/30s steepeners despite locally steep levels. Thanks to Dr. Yellen, the yield curve is gellin'.	- Dr. Yellen is Gellin' - TIPS – What Are They Waiting For? - Short Duration: Belly Remains Attractive - Hedging a Threshold Shift: Enter Bull Steepeners - Agency MBS: Overweight Belly Download Report
Stopped Out	November 8, 2013
When the facts change, we change. Higher-than-expected 3Q13 real GDP growth and a higher-than-expected nonfarm payroll report increase the odds of an earlier-than-expected taper. We move to neutral on duration and on the shape of the yield curve. We see 10-year yields trading in a range of 2.60-2.90% into the December FOMC meeting.	- She Blinded Me with Dismal Science - Re-Evaluating Trading Strategies - Watch the Thresholds! - Supply Outlook - Momentum Shift Bullish for Vol - Chock Full of Surprises Download Report
Every Rose Has Its Thorn	November 1, 2013
We remain bullish on the Treasury market, but less so relative to just prior to the October FOMC meeting. We still believe 10-year yields will trade into a 2.20-2.40% range, but recognize that investors will require softer data to justify those yield levels now. We suggest buying dips in 10-year yields in the 2.60-2.65% range.	- Tracking Term Premiums - Pedal Off the Metal - Exploiting Potential Regime Shifts - The Seasonal Inflation Curse - Tracking the Flows - Daydream or Nightmare? - Changing Our View - On the Edge, Again Download Report

Event Calendar

Monday	Tuesday	Wednesday	Thursday	Friday
23-Dec	24-Dec	25-Dec	26-Dec	27-Dec
US: U Mich. Confidence	US: MBA Mortgage Applications US: New Home Sales FR: GDP	Christmas Day JPY: BoJ Nov minutes	US: Initial Jobless Claims JPY: Jobless Rate JPY: CPI JPY: Industrial Production	
30-Dec	31-Dec	1-Jan	2-Jan	3-Jan
SP: Retail Sales UK: Nationwide House Prices		New Year's Day SP: Unemployment Rate	US: Initial Jobless Claims US: ISM Manufacturing EU: PMI Manufacturing	US: Fed's Plosser and Bernanke Speak

Source: Morgan Stanley Research

Government Bond Supply

23-Dec	24-Dec	25-Dec	26-Dec	27-Dec
US: 1m T-Bill \$20bn, 3m T-Bill \$32bn, 6m T-Bill \$27bn ITA: BTP Announcement		JPY: 2y JGB, ¥2900bn*	US: 1m, 3m, 6m T-Bill Announcement	ITA: CTZ Auction Tap, €2.5bn* (Possible CTZ December 2015)
US OMO: 22-30y T, \$1.25 - \$1.75				
30-Dec	31-Dec	1-Jan	2-Jan	3-Jan
US: 1m T-Bill \$20bn*, 3m T-Bill \$32bn*, 6m T-Bill \$27bn* ITA: BTP Auction, €2.5bn*			US: 3m, 6m T-Bill Announcement US: 1y T-Bill Announcement US: New 3y, Reopening 10y and 30y UST Announcement	

Source: National Treasuries, Federal Reserve, Morgan Stanley Research

Note: * = Size Estimated; ** = Auction not announced yet; *** Syndication/Mini Tender will happen in the week commencing; T: Treasury Purchase, TII: TIPS Purchase, S: Treasury Sales

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Treasury Yield Forecasts

Base Case							
Quarter	Exp*	2y	3y	5y	7y	10y	30y
3Q13a	-	0.37	0.70	1.50	2.13	2.74	3.75
4Q13	+1.5m	0.30	0.60	1.35	2.05	2.70	3.75
1Q14	-1.5m	0.55	0.90	1.65	2.35	3.05	4.10
2Q14	-6m	0.80	1.15	1.95	2.60	3.25	4.15
3Q14	-3m	1.00	1.40	2.20	2.80	3.40	4.20
4Q14	-	1.15	1.55	2.30	2.90	3.45	4.25

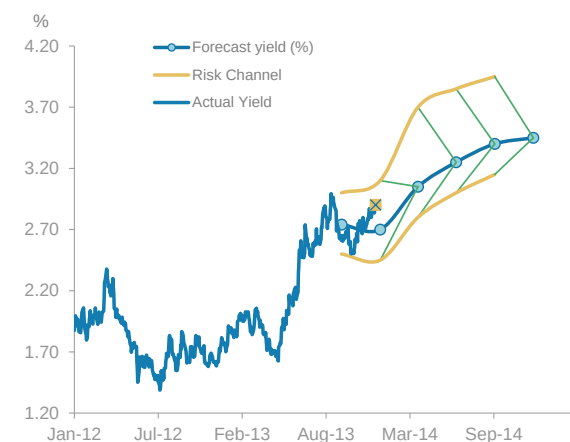
Bull Case							
Quarter	Exp*	2y	3y	5y	7y	10y	30y
3Q13a	-	0.37	0.70	1.50	2.13	2.74	3.75
4Q13	+3m	0.25	0.50	1.25	1.90	2.60	3.65
1Q14	+6m	0.20	0.35	1.05	1.65	2.30	3.30
2Q14	+7.5m	0.20	0.30	0.95	1.55	2.25	3.30
3Q14	+4.5m	0.25	0.35	0.95	1.55	2.20	3.30
4Q14	-	0.40	0.45	1.05	1.65	2.30	3.30

Bear Case							
Quarter	Exp*	2y	3y	5y	7y	10y	30y
3Q13a	-	0.37	0.70	1.50	2.13	2.74	3.75
4Q13	-1.5m	0.45	0.80	1.60	2.25	2.95	4.00
1Q14	-6m	0.95	1.35	2.20	2.90	3.60	4.65
2Q14	-6m	1.25	1.65	2.50	3.20	3.80	4.70
3Q14	-3m	1.50	1.90	2.75	3.40	3.95	4.80
4Q14	-	1.65	2.10	2.90	3.50	4.05	4.80

Base Case Risk Channel							
Quarter	Exp*	2y	3y	5y	7y	10y	30y
Current	-	0.29	0.45	1.16	1.81	2.46	3.51
Current	-	0.54	0.85	1.70	2.39	3.09	4.13
4Q13	+1.5m	0.30	0.45	1.15	1.80	2.45	3.50
4Q13	+1.5m	0.55	0.85	1.70	2.40	3.10	4.15
1Q14	-1.5m	0.50	0.70	1.45	2.15	2.80	3.85
1Q14	-1.5m	1.00	1.45	2.30	3.00	3.70	4.75
2Q14	-6m	0.70	0.95	1.75	2.40	3.00	3.90
2Q14	-6m	1.25	1.65	2.50	3.20	3.85	4.80
3Q14	-3m	0.90	1.20	1.95	2.55	3.15	3.95
3Q14	-3m	1.35	1.80	2.65	3.30	3.95	4.85
4Q14	-	1.05	1.30	2.05	2.65	3.20	4.00
4Q14	-	1.55	1.95	2.80	3.40	4.00	4.85

Maturity	Current Market	4Q13			1Q14		
		Forecast	Market Forward	Forecast less Fwd	Forecast	Market Forward	Forecast less Fwd
Yield	(%)	(%)	(%)	(bp)	(%)	(%)	(bp)
2-Year	0.344	0.300	0.354	-5.4	0.550	0.480	7.0
3-Year	0.743	0.600	0.758	-15.8	0.900	0.941	-4.1
5-Year	1.635	1.350	1.651	-30.1	1.650	1.829	-17.9
7-Year	2.301	2.050	2.315	-26.5	2.350	2.467	-11.7
10-Year	2.927	2.700	2.936	-23.6	3.050	3.054	-0.4
30-Year	3.853	3.750	3.859	-10.9	4.100	3.915	18.5
Curve	(%)	(%)	(%)	(bp)	(%)	(%)	(bp)
2s/3s	0.399	0.300	0.403	-10.3	0.350	0.461	-11.1
2s/5s	1.291	1.050	1.296	-24.6	1.100	1.349	-24.9
2s/7s	1.957	1.750	1.961	-21.1	1.800	1.987	-18.7
2s/10s	2.583	2.400	2.581	-18.1	2.500	2.574	-7.4
2s/30s	3.509	3.450	3.504	-5.4	3.550	3.434	11.6
3s/5s	0.892	0.750	0.893	-14.3	0.750	0.888	-13.8
3s/7s	1.558	1.450	1.557	-10.7	1.450	1.526	-7.6
3s/10s	2.184	2.100	2.178	-7.8	2.150	2.113	3.7
3s/30s	3.110	3.150	3.101	4.9	3.200	2.973	22.7
5s/7s	0.666	0.700	0.664	3.6	0.700	0.638	6.2
5s/10s	1.292	1.350	1.285	6.5	1.400	1.225	17.5
5s/30s	2.218	2.400	2.208	19.2	2.450	2.085	36.5
7s/10s	0.626	0.650	0.621	2.9	0.700	0.587	11.3
7s/30s	1.552	1.700	1.544	15.6	1.750	1.447	30.3
10s/30s	0.926	1.050	0.923	12.7	1.050	0.860	19.0
Butterfly	(%)	(%)	(%)	(bp)	(%)	(%)	(bp)
2s/3s/5s	-0.494	-0.450	-0.490	4.0	-0.400	-0.427	2.7
2s/5s/7s	0.625	0.350	0.632	-28.2	0.400	0.712	-31.2
2s/5s/10s	-0.001	-0.300	0.011	-31.1	-0.300	0.124	-42.4
2s/5s/30s	-0.927	-1.350	-0.912	-43.8	-1.350	-0.736	-61.4
2s/7s/10s	1.331	1.100	1.340	-24.0	1.100	1.400	-30.0
2s/7s/30s	0.405	0.050	0.417	-36.7	0.050	0.540	-49.0
2s/10s/30s	1.657	1.350	1.659	-30.9	1.450	1.714	-26.4
3s/5s/7s	0.227	0.050	0.228	-17.8	0.050	0.250	-20.0
3s/5s/10s	-0.399	-0.600	-0.392	-20.8	-0.650	-0.337	-31.3
3s/10s/30s	1.258	1.050	1.255	-20.5	1.100	1.253	-15.3
5s/7s/10s	0.040	0.050	0.044	0.6	0.000	0.050	-5.0
5s/7s/30s	-0.886	-1.000	-0.879	-12.1	-1.050	-0.810	-24.0
5s/10s/30s	0.366	0.300	0.362	-6.2	0.350	0.365	-1.5
7s/10s/30s	-0.300	-0.400	-0.302	-9.8	-0.350	-0.273	-7.7

10-Year Base Case Forecast | Realized Yield | Risk Channel

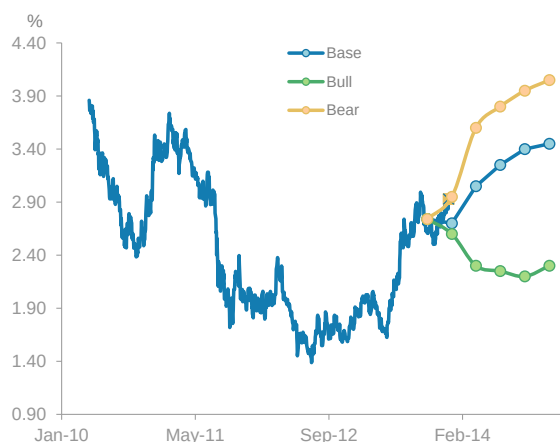


Source: Morgan Stanley Research

Updated on: 20, Dec 2013 * Expectations refer to incremental extension or retraction of the market-implied fed funds target rate

For more information on our rate forecast methodology please consult [US Interest Rate Strategist: Unconventional Uncertainty](#).

10-Year Bull | Bear | Base Case Scenarios



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US Governments

The English Patient

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- As the unemployment rate moves from above 6.5% to below 6.5%, the credibility of the Fed's rate guidance naturally weakens due to the time inconsistency problem. Because the FOMC suggested moving the unemployment rate threshold was off-the-table for now, the Treasury market has begun to price in a less credible rate guidance framework earlier than we expected prior to the meeting.
- Given the uncertainty around what determines the Fed's collective view once the unemployment rate threshold is crossed, the market is left to judge today's rate guidance on its credibility – i.e., the probability the Fed will not renege on today's rate guidance tomorrow.
- The rate path priced by the market will require a higher level of term premium – a dynamic that is likely to manifest most strongly in the 5- to 7-year sector. We suggest going short the 7-year point on the 3s/7s/10s butterfly (50:50 weighted), or going short the 5-year point on the 2s/5s/30s butterfly (50:50 weighted).
- For a less directional play that retains the potential to profit from lower rate guidance credibility, we suggest entering a 3s/5s curve flattener versus a 10s/30s curve flattener (1.0:0.5 DV01 weights of curves).

Toward A Less Credible Rate Guidance Framework

As the unemployment rate moves from above 6.5% to below 6.5%, the credibility of the Fed's rate guidance naturally weakens due to the time inconsistency problem. Because the FOMC suggested moving the unemployment rate threshold (URT) was off-the-table for now, the Treasury market has begun to price in a less credible rate guidance framework earlier than we expected prior to the meeting. In our initial reaction piece to the FOMC announcement¹, we explained our view that the FOMC tried to guide the market away from relying on the URT. Instead, Bernanke emphasized that markets should soon begin to focus on a broader range of data (including the inflation rate) in order to judge how long policy will be on hold.

We explained that the market's best guide – at this point – to assess the length of this period is to look at the Fed's collective assessment of the appropriate pace of policy tightening. But as Bernanke suggested, even the "blue dots"

in the Summary of Economic Projections (SEP) only give "some sense" of the collective view:

The SEP, the Survey of Economic Projections, which were distributed, obviously that's individual assessments and not the committee's collective view. But nevertheless, it gives you some sense of current expectations about the length of time.

Given the uncertainty around what determines the collective view once the URT is crossed, **the market is left to judge today's rate guidance on its credibility – i.e., the probability the Fed will not renege on today's rate guidance tomorrow.** The implementation of the threshold framework was meant to reduce the probability that the market would question the credibility of the Fed's rate guidance. Now that the Fed is trying to divert the market's attention away from the URT – thereby allowing the threshold framework to lapse – the market has more freedom to question the credibility of the Fed's rate guidance.

What is the Time Inconsistency Problem?

The time inconsistency problem was most recently detailed in a paper by English et al² – a seminal work on the threshold-based rate guidance framework, in our view. The problem arises when the incentives faced by policy makers today are inconsistent with those in the future. For the Fed, the low rate regime is the optimal policy path now, because low rates will provide the accommodation necessary to bring the economy back to potential. However, the future costs – overshooting inflation and output objectives – will shift the future optimal policy path and provide the Fed with a strong incentive to later renege on its past commitments. If forward-looking markets understand the Fed's future incentives, then the markets will never price in the optimal path today, and this produces sub-optimal outcomes in the future.

As a result, the success of monetary policy is largely dependent on success of the communication strategy and the perceived credibility of central banker commitments. In light of this, the threshold-based rate guidance framework was designed to be durable enough to ensure that the Fed remains true to its promises despite the strong incentives to renege in the future.

¹ See [US Interest Rate Strategy Insight: Our Reaction to the FOMC](#), December 19, 2013

² See English et al [The Federal Reserve's Framework for Monetary Policy – Recent Changes and New Questions](#), November 7-8, 2013

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The Keymaster and The Gatekeeper³

Of course, the credibility of forward rate guidance relies on more than just how the framework handles the time inconsistency problem. Credibility also depends on who is dictating the framework, and how they may change it over time. This is why the idea of transition between Fed Chairs can cause term premiums to rise – as it did earlier this year. The transition between Chairman Bernanke (the Gatekeeper) and next Chairperson (the Keymaster) introduced uncertainty about how the threshold-based rate guidance framework would change, if at all. When the market suspected that Larry Summers would guide monetary policy under a different framework, Treasury yields soared because the market assigned less credibility to the threshold-based guidance.

However, when Larry Summers withdrew from the pool of candidates, the market restored the framework's credibility because it expected (1) Janet Yellen would become the next Fed Chair, (2) she would favor an optimal control-like policy, and (3) she would lower the URT to enhance the credibility of the framework that best achieved an optimal control-like outcome. Indeed, according to the English et al simulations, the current rate guidance framework (6.5% unemployment and 2.5% inflation projection threshold) will produce sub-optimal outcomes. The market reasoned that Yellen would use this work as justification to, at some point, lower the URT – effectively prolonging the existence of the threshold-based rate guidance framework, and increasing the credibility of the current rate guidance itself.

However, because the Fed shied away from lowering the URT at the December meeting, the market should believe that the more credible threshold-based framework will soon give way to a less credible form of rate guidance. **Exhibit 1** highlights the scenarios for when the UR would reach 6.5% based on the average increase in nonfarm payroll employment and the labor force participation rate (LFPR) at the time. If the LFPR returns to its cyclical low of 62.8% and nonfarm payrolls continue to increase at a pace of 200,000 jobs per month, we would expect the UR to reach 6.5% at the March nonfarm payroll report (released in early April 2015). The green cells represent the risk scenarios for the UR hitting the 6.5% threshold earlier-than-projected by the FOMC's central tendency range. The yellow cells suggest the scenarios assumed by the central tendency range.

With the unemployment rate (UR) nearing 6.5%, the English patient – i.e., threshold-based guidance – is on its deathbed unless the FOMC administers a different type of medicine.

³ <http://www.ghostbusters.com/>

Exhibit 1

Timeline to Reach 6.5% Unemployment⁴ under Various Participation Rates

Participation rate	Average Total NFP change per month ('000)				
	150	175	200	225	250
62.7%	Apr-14	Mar-14	Feb-14	Feb-14	Feb-14
62.8% (Recent low)	Sep-14	Jun-14	Apr-14	Apr-14	Mar-14
62.9%	Feb-15	Sep-14	Jul-14	May-14	May-14
63.0% (Current)	Jul-15	Dec-14	Sep-14	Jul-14	Jun-14
63.1%	Dec-15	Mar-15	Nov-14	Sep-14	Aug-14
63.2%	May-16	Jun-15	Feb-15	Nov-14	Sep-14
63.3%	Oct-16	Oct-15	Apr-15	Jan-15	Nov-14

Source: Morgan Stanley Research, Bureau of Labor Statistics

And unfortunately for Treasury market investors, Bernanke hinted that the medicine may not change in the very near term – even under the new doctor. At the press conference, when asked if the current policy would continue under Yellen's leadership, Bernanke said:

Yes, it will. I have always consulted closely with Janet, even well before she was named by the president, and I consulted closely with her on these decisions, as well, and she fully supports what we did today.

This makes it much less likely that Yellen, if confirmed, will change course immediately. As a result, Treasury market investors should assume the transition to a less credible rate guidance framework will continue. That is, of course, unless (1) the Minutes of the December FOMC meeting suggest participants would consider a lower URT or an inflation floor at some point soon, (2) the UR starts moving higher (and further away from 6.5%), or (3) PCE inflation continues to fall toward zero.

And When Rate Guidance Loses Credibility?

Investors get nervous about the future policy path. This is even more the case when the central bank providing the rate guidance sees the risks to the outlook for the economy and the labor market as more balanced than they have been in years. In fact, the December FOMC statement was the first time since the July 2011 FOMC statement that the Fed did not mention “downside risks” when referring to the outlook (see **Exhibit 2**).

The Committee sees the risks to the outlook for the economy and the labor market as having become more nearly balanced.

As a result, the rate path priced by the market will require a higher level of term premium – a dynamic that is likely to

⁴ Assumes a population growth rate equal to the average rate over the past 12 months. The months in the table represent the months in which the labor report would be expected to project a 6.5% unemployment rate.

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manifest most strongly in the 5- to 7-year sector. Thus, we target selling Treasuries in this sector to maximize returns, as the market begins to price in the shortfall of credibility in the revised rate guidance. A second factor that favors shorts in the 5- to 7-year sector is MBS tapering. We believe investors who sell MBS to the Fed eventually replace them, at least partially, with Treasuries in the 5- to 7-year sector (as a result of scant net issuance in agency MBS). Thus, to the extent that continued tapering leads to a rise in term premiums, the 5- to 7-year sector should continue to cheapen.

Optimal Ways to Go Short

We prefer expressing our short bias in the 5- to 7-year sector by going short the 7-year point on the 3s/7s/10s butterfly (50:50 weighted), or going short the 5-year point on the 2s/5s/30s butterfly (50:50 weighted).

To look for the optimal position to be short, we look for butterfly structures that have been well correlated to the rates levels and (1) offer a lower negative risk-adjusted rolldown and carry and (2) look fair to rich versus outright level of rates. In our screen in **Exhibit 3** we can see that the 3s/7s/10s butterfly offers both these advantages versus being outright short the 7-year sector. The trade loses 4.7bp in rolldown and carry in three months, lower than other butterflies in our screen. **Exhibit 4** shows the five year history of the butterfly structure, and our target level.

Lower-Risk Ways for Positioning Short

The trade we have described above is likely to profit if the market continues to assign low credibility to the current rate guidance. However, the risk to the trade is that incoming economic data disappoints meaningfully, or the Fed takes concrete steps to add more credibility to the rate guidance, thereby leading the market to rally. For investors wishing to hedge this risk, while also retaining some potential to profit from the lowering credibility of rate guidance, we suggest:

- 3s/5s Flatteners versus 10s/30s Flatteners (DV01 weights of curves = 1.0:0.5).

The trade is weighted based on the regression coefficient of the 3s/5s curve versus the 10s/30s curve over the past six months. The trade offers slightly negative rolldown and carry of 0.6bp over three months. In our view, the trade can profit under the following scenarios:

Exhibit 2

FOMC Statement: Risks to the Outlook

Date	Description of Risks to the Outlook	10y Yield* (%)
06/22/2011	No risk statement included	3.01
08/09/2011	"downside risks ... have increased"	2.20
09/21/2011	"significant downside risks"	1.88
...	...	...
12/12/2012	"significant downside risks"	1.72
01/30/2013	"downside risks"	2.03
03/20/2013	"downside risks"	1.96
05/01/2013	"downside risks"	1.66
06/19/2013	"downside risks ... having diminished"	2.33
07/31/2013	"downside risks ... having diminished"	2.57
09/18/2013	"downside risks... diminished, but the tightening of financial conditions..."	2.66
10/30/2013	"downside risks ... having diminished"	2.53
12/18/2013	"risks to the outlook ... more nearly balanced"	2.89

Source: Morgan Stanley Research, Federal Reserve, * Close of FOMC announcement day

Exhibit 3

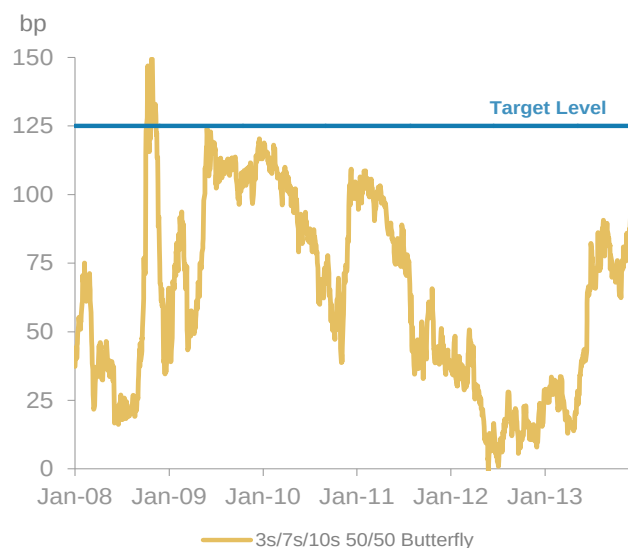
Best Butterflies to Position Short

Butterfly	3m Z-Score	3y Z-Score	3m R ²	Fly 3m RD&C (bp)	Fly 3m RD&C Quot	Fly RD&C Adv	Fly Resid (bp)	3m Z-Score
2/5/10	3.1	1.3	66%	14.4	64%	0%	4.6	1.35
2/7/10	2.6	1.5	97%	11.8	40%	10%	2.3	1.40
3/7/30	2.7	1.7	96%	12.2	44%	7%	3.4	1.60
2/7/30	2.8	1.6	96%	19.2	55%	-4%	5.1	2.14
3/5/30	3.0	1.5	61%	14.8	64%	0%	5.2	1.29
2/5/30	2.9	1.5	66%	21.8	71%	-8%	5.1	1.04
3/7/10	2.4	1.5	90%	4.7	21%	30%	0.6	0.22
2/10/30	2.1	1.4	97%	13.0	41%	1%	1.3	0.72
5/7/10	2.3	2.2	80%	1.7	19%	32%	0.4	0.27
2/5/10	3.1	1.3	66%	14.4	64%	0%	4.6	1.35

* RD&C = Rolldown and Carry, RD&C Quot. = RD&C Quotient, Fly RD&C Adv = Advantage in risk adjusted rolldown and carry in holding long in the belly versus long in the butterfly
Source: Morgan Stanley Research

Exhibit 4

3s/7s/10s 50:50 Butterfly



Source: Morgan Stanley Research, Federal Reserve

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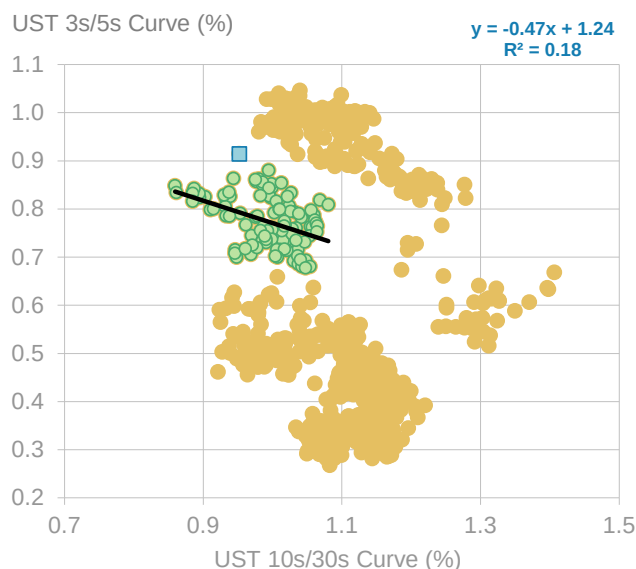
Scenario 1 - Market continues to sell off: If economic data continues to maintain its recent positive tone, then the markets could start pushing further against the Fed's rate guidance. In such a scenario, we would expect 3s to start cheapening up more than recently, as they unhook from the blue dot levels that the FOMC members submitted as their rate projections in the Summary of Economic Projections (SEP) at the December FOMC meeting. There is precedent for 3s to have led the cheapening in Treasuries – right before the September FOMC when the market assigned very low credibility to the Fed's verbal guidance for separating tapering from tightening. Such a cheapening of 3s could start flattening the 3s/5s curve or keep it flat, while the 10s/30s curve is likely to continue to flatten as the market maintains its hawkish bias.

Scenario 2 - Market rallies: If the economic data disappoints or the Fed takes concrete steps to reinforce its rate guidance, we would expect the 3s/5s curve to bull flatten, while the 10s/30s curve could bull steepen. Based on our regression weights, we would expect the two effects to cancel each other out. However, given the high residual of the 3s/5s curve's current level versus the 10s/30s curve, we could see a correction, which would be profitable for the trade (see **Exhibit 5**).

The risk to the trade is that the 3s/5s curve steepens at a much faster pace than the 10s/30s flattening pace implied by our regression weights.

Exhibit 5

3s/5s Curve vs. 10s/30s Curve (Last Three Years)



Source: Morgan Stanley Research

Appendix

What Are Time Inconsistencies?

In plain English, dynamic inconsistency describes a situation where policy makers' preferences change over time, and if acted on could cause sub-optimal outcomes. In the current environment, policy makers have an incentive to commit to keeping rates low; however, in the future when the economic situation improves – inflation and unemployment are near to their respective targets – they will have an incentive to renege on their commitments. To illustrate the idea, we create a stylized model. In this model, we assume central bankers want to maximize their happiness or “utility” and their utility “u” is maximized when inflation equals the long-run goal of 2% and unemployment equals the natural rate of unemployment. Below, we use the simple equation for a circle to model the central banker utility maximization problem, where U and U* are unemployment and the natural rate of unemployment respectively and π is the inflation rate.

$$\text{Max } [u = -0.5[(\pi - 2\%)^2 + (U - U^*)^2]]$$

Meanwhile, we assume that inflation is a function of inflation expectations and the employment gap. In this traditional short run supply Phillips curve, the Fed influences movements along the curve by changing money supply and influences shifts in the curve by managing inflation expectations. The traditional Phillips curve relationship can be modeled linearly as follows:

$$\Delta P = \Delta P^e - \theta(U - U^*)$$

Where ΔP is inflation, ΔP^e is inflation expectations and U and U* are unemployment and the natural level of unemployment, respectively.

Exhibit 1A plots these functions on an inflation and unemployment plane. The t=1 “dot” is the current level of the employment gap and inflation and we assume that inflation expectations are anchored at the Fed’s 2% long-term target. We use a simple linear regression to estimate the current Phillip curve slope. In this illustration, the t=1 “dot” is on the utility curve “u2,” implying that the Fed has the incentive to provide accommodation to reach higher levels of utility. In the current zero lower bounded environment, the Fed provides accommodation through its commitment to keep rates on hold for longer than conventional policy

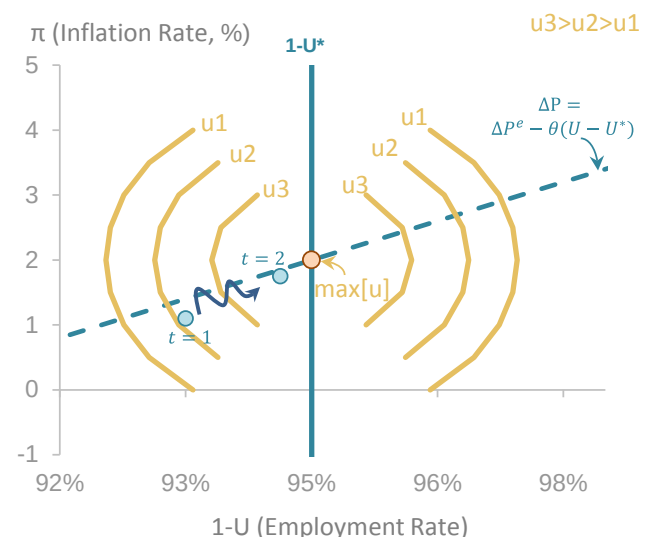
rules would imply. This accommodation moves the economy to the t=2 “dot” and where the Fed is closer to its utility maximization point, and hence, has less incentive to provide the accommodation it promised. If forward-looking markets understand these changing preferences, they will never price real rates low enough to allow the optimal accommodation to be achieved. Alan Blinder, former vice-chairman of the Federal Reserve, suggests the following when discussing central bank credibility:

In a game-theoretic setting, credibility is identified with either dynamic consistency or incentive compatibility...

The rate guidance framework created the incentive compatibility for the policy. Within this framework, the success of monetary policy is largely dependent on success of the communication strategy and the perceived credibility of central banker commitments. In light of this, the rate guidance framework was designed to be durable enough to ensure that the Fed remains true to its promises despite the strong incentives to renege in the future. Still, according to the English et al. simulations, the current rate guidance framework (6.5% unemployment and 2.5% inflation projection threshold) will produce sub-optimal outcomes. Hence, absent lowering the URT, the Fed must rely on verbal guidance about its behavior after the thresholds are crossed.

Exhibit 1A

Time Inconsistency Illustration



Source: Morgan Stanley Research

US Interest Rates

Our Reaction to the FOMC

Matthew Hornbach and team (212) 761-1837

- We turn neutral on yield curve shape and remain neutral on duration. Treasury yields could remain under upward pressure as long as growth data evolve in line with the Fed's projections and PCE inflation begins to move higher.
- As yields move higher, the yield curve should bear-flatten led by the 5-year maturity point. We suggest investors remain long implied rate volatility.

Angels We Have Heard on High

The message the Fed tried to deliver to the bond market was that the level of accommodation it is now providing to the economy is "more or less" the same as before, but that it is gradually moving away from its supplementary easing measures (LSAPs). By the end of the trading day, 10-year Treasury yields reacted to this message by remaining more or less unchanged from levels that existed in the morning before the announcement. At the press conference, Bernanke made it clear that,

the action today is intended to keep the level of accommodation more or less the same overall and enough to push the economy forward.

The Fed decided to taper \$5bn UST to \$40bn/month, and taper \$5bn MBS to \$35bn/month. The Fed also strengthened its forward rate guidance by saying it will be on hold "well past" a decline in the unemployment rate (UR) below 6.5%, "especially if projected inflation continues to run below the Committee's 2 percent longer-run goal." Bernanke also mentioned this at the press conference:

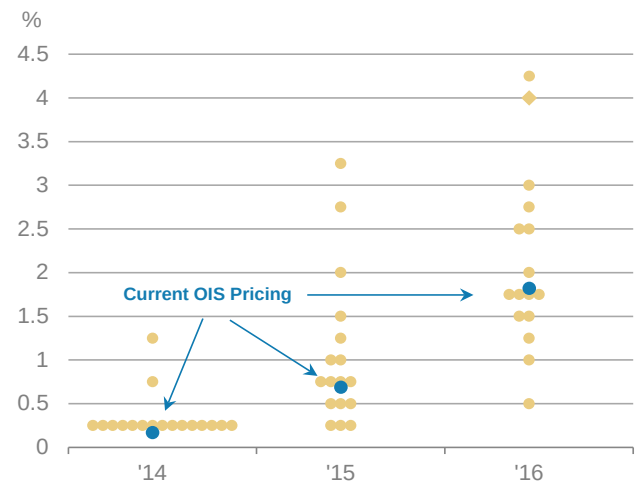
we strengthened our guidance to make clear that we expect to keep rates low well beyond the point that unemployment hits 6.5 percent.

So, as long as the Fed projects inflation to remain below its 2% goal (which it is doing currently through 2016 with both the core and headline PCE), the Fed will remain on hold "well past" the time when the UR falls below the 6.5% unemployment rate threshold (URT). This "well past" language was in addition to the previous "at least as long as" language – which the Fed kept in the official statement.

In addition, the Fed's forward rate guidance became more accommodative on the margin because some participants lowered their expectations of the appropriate pace of policy firming – i.e., the 'blue dots'.

Exhibit 1

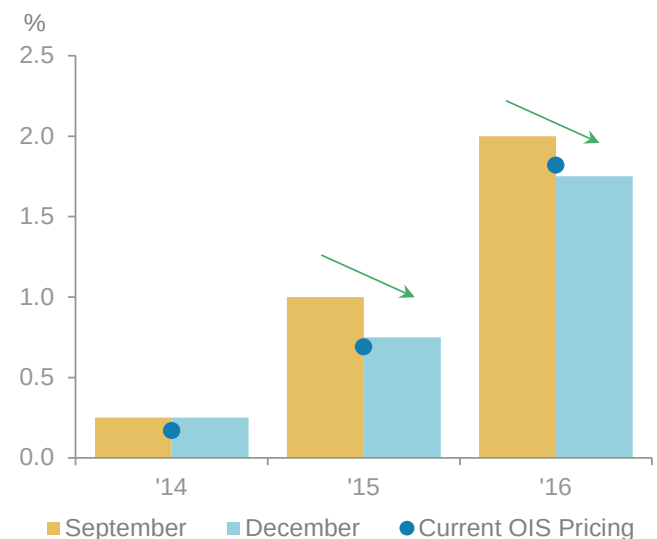
Appropriate Pace of Policy Firming ('Blue Dots') vs. Current OIS Market Pricing



Source: Morgan Stanley Research, Federal Reserve

Exhibit 2

Change in Median/Modal Pace of Policy Firming and Current OIS Market Pricing



Source: Morgan Stanley Research, Federal Reserve

Exhibit 1 shows the current set of policy paths expected by FOMC participants and market expectations priced into the overnight index swap market (OIS). **Exhibit 2** shows how the median (and coincidentally modal) path for the fed funds target rate fell from the September to the December Summary of Economic Projections (SEP). In effect, the Fed 'cut' the

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pace of expected policy tightening by 25bp between 2014 and 2015.

When Is The Next Taper?

Now that the Fed has announced tapering in a 'modest' way, subsequent decisions on further tapering will be driven by economic data. Our framework for assessing the risks around our economist's views on tapering has rested on two pillars in the past: (1) the degree of labor market improvement, and (2) the degree of the Fed's confidence in continued improvement. Speeches by New York Fed President Dudley and Boston Fed President Rosengren have suggested that the Fed's confidence is determined by the real growth rate of the economy. Interestingly, Rosengren dissented because he was not confident enough in the sustainability of the growth data, let alone the employment data.

At the September FOMC meeting, real growth through the first three quarters of the year was near 2.0% annualized without any signs of acceleration. Without the acceleration in growth and the prospects for lower growth from tighter financial conditions and fiscal policy risks, the Fed lacked confidence in the outlook for the labor market and delayed tapering. However, with 4Q13 real GDP tracking close to 2.0% heading into the December meeting, full-year real GDP was annualizing near 2.3% – at the upper bound of the Fed's 2013 central tendency range – with some signs of acceleration into 2014. This, along with lessened concern over fiscal policy risks, provided the Fed with enough confidence to justify a 'modest' taper.

Going forward, incoming data will have to meet Fed expectations for growth and employment, but also possibly inflation at some point later on the tapering timeline. In the press conference, Bernanke suggested,

If the incoming data broadly support the Committee's outlook for employment and inflation, we will likely reduce the pace of securities purchases in further measured steps at future meetings.

It could be the Fed will only require a continued acceleration in real growth to justify tapering at the next several FOMC meetings. If real growth gave the Fed confidence in the future of the labor market, perhaps it would also engender confidence in higher inflation. But if inflation does not begin to move toward the Fed's 2% goal, the Fed could change course.

How Does the Taper Impact Net Treasury Supply?

If we assume the forecasts of our US economists hold throughout 2014, we believe the Fed could taper \$10bn at each FOMC meeting until the October 28-29 meeting. After all, the forecasts of our US economists are similar in many ways to those of the FOMC's central tendency range. In addition, in the press conference, Bernanke suggested that the Fed could do a "measured reduction" in its asset purchases at each meeting in 2014 if the economy continues to "make progress" in terms of job gains and inflation:

If we're making progress in terms of inflation and continued job gains, then I imagine we'll continue to do probably at each meeting a measured reduction. That would take us to late in the year.

Exhibit 3 shows our expected net issuance in Treasury coupons projected for each month in 2014, assuming the Fed tapers \$10bn (\$5bn in Treasuries) at each of the upcoming FOMC meetings. The total Fed purchases in 2014 under this scenario would be \$255bn, almost \$100bn lower than our initial estimate of \$360bn (we had assumed tapering to be announced at every other meeting).

Exhibit 3

Expected Monthly Net Coupon Issuance in 2014 Assuming No Coupon Size Cuts

Month	Redemptions	2s	3s	5s	7s	10s	30s	Total Coupons	Total TIPS	5y TII	10y TII	30y TII	Net Supply	Fed Purchases	Net Supply Ex-Fed
Dec-14	111	32	30	35	29	21	13	160	8	8			57	0	57
Nov-14	138	32	30	35	29	24	16	166	13		13		41	0	41
Oct-14	109	32	30	35	29	21	13	160	17	10		7	68	10	58
Sep-14	108	32	30	35	29	21	13	160	13		13		65	15	50
Aug-14	133	32	30	35	29	24	16	166	8	8			41	15	26
Jul-14	130	32	30	35	29	21	13	160	15		15		45	20	25
Jun-14	105	32	30	35	29	21	13	160	15	8		7	70	25	45
May-14	130	32	30	35	29	24	16	166	13		13		49	30	19
Apr-14	122	32	30	35	29	21	13	160	10	10			48	30	18
Mar-14	103	32	30	35	29	21	13	160	13		13		70	35	35
Feb-14	130	32	30	35	29	24	16	166	9			9	45	35	10
Jan-14	126	32	30	35	29	21	13	160	15		15		49	40	9
Total	1448	384	360	420	348	264	168	1944	149	44	82	23	645	255	390

*Redemptions include TIPS and coupons

Source: Morgan Stanley Research, Federal Reserve

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Turning to the December announcement, **Exhibit 4** shows how the net supply in Treasuries and MBS will look, given the pace of purchases in Treasuries and agency MBS starting in January. At the new pace, the Fed will continue to remove almost half the 10-year duration equivalents added to the market via net issuance in Treasuries and MBS, respectively. Continued tapering at a smaller, similar or larger size will accordingly increase or decrease the net issuance removed by the Fed each month.

A \$5bn decrease in Fed purchases in Treasuries, given the current allocations, would be equivalent to a \$5.5bn decrease in 10-year duration equivalents. Similarly, a \$5bn decrease in par MBS purchases would be equivalent to a \$3bn decrease in 10-year duration equivalents. For our calculations, we assume that investors that sell MBS to the Fed, and eventually replace them by purchasing Treasuries in the 4.75- to 7-year sector.

The Future of the Thresholds

Our interpretation of the statement is that the Fed wants to lessen the focus on the unemployment rate threshold (URT) by making it into a marker of sorts, and increase the focus on inflation (or lack thereof). Bernanke mentioned in the press conference that the Fed had no immediate plans to lower the URT "in the very near term". In addition, Bernanke suggested the Fed was comfortable

setting a 6.5 percent unemployment rate as the point at which we would begin to look at a more broad set of labor market indicators.

However, precisely because we don't want to look just at the unemployment rate, we want to – once we get to 6.5 percent, we want to look at hiring, quits, vacancies, participation, long-term unemployment, et cetera, wages,

we couldn't put it in terms of another unemployment rate level, specifically.

So I expect there will be some time past the 6.5 percent before all of the other variables we'll be looking at will line up in a way that will give us confidence that the labor market is strong enough to withstand the beginning of increases in rates.

At the same time, the Committee left the door open for the implementation of an inflation floor and possibly a cut in the URT. At the press conference, Bernanke reaffirmed this by saying,

And while the view of the Committee was that the best way forward today was in this more qualitative approach, which incorporates elements both of the unemployment threshold and the inflation floor, that further strengthening would be possible and it's something that has certainly not been ruled out.

So at the same time the Fed 'strengthened' its rate guidance with respect to inflation, it also began to steer the market away from using the UR to determine when policy might tighten and toward using a broader range data. The FOMC statement makes this clear:

In determining how long to maintain a highly accommodative stance of monetary policy, the Committee will also consider other information, including additional measures of labor market conditions, indicators of inflation pressures and inflation expectations, and readings on financial developments.

That broader range of data will determine the length of the on-hold period, which the FOMC statement describes as lasting "well past" the 6.5% crossing.

Exhibit 4

Net Issuance in Treasury and MBS Markets Under Current Tapering*** Scenario

Sector	Percent Allocation*	Par Amount (\$ bn)			10-Year Equivalents (\$ bn)		
		Baseline Issuance (Net)**	Fed Purchase (35 MBS/40 Tsy)	Supply After Fed Impact	Baseline Issuance (Net)**	Fed Purchase (35 MBS/40 Tsy)	Supply After Fed Impact
0-2y	0%	-89	0	-89	0	0	0
2-4y	0%	30	0	30	11	0	11
4-4.75y	11%	0	4	-4	0	2	-2
4.75-5.75y MBS	12%	41	22	19	25	13	12
5.75-7y MBS	16%	35	24	11	28	15	13
7-10y	29%	22	12	10	22	10	12
10-20y	2%	0	1	-1	0	1	-1
20-30y	27%	14	11	3	29	21	7
TIPS	3%	12	1	11	17	2	15
Total Purchases	100%	66	75	-9	131	65	66

* Percent allocation only applicable to US Treasuries. ** The baseline issuance assumes continuation of the current coupon auction sizes each month, and the average monthly pace of Treasury coupon redemptions expected in 2014. *** A \$5bn decrease in Fed purchases in Treasuries, assuming the current allocation, would be equivalent to a \$5.5bn decrease in 10-year duration equivalents. Similarly, a \$5bn each decrease in par MBS purchases would be equivalent to a \$3bn decrease in 10-year duration equivalents.

Source: Morgan Stanley Research estimates, Federal Reserve

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Inflation Is the Cost of Keeping Rates on Hold

We believe the Committee is shifting market focus away from the unemployment rate threshold and toward inflation. This is appropriate because the Committee needs to see inflation starting to tick higher before it is comfortable raising rates. It's no secret that the Phillips curve – the relationship between unemployment and inflation – has been flattening since the 1970 US hyper-inflation. **Exhibit 5** plots the *Phillips* curve since the late 1980s. As the exhibit illustrates, the slope of the curve has been falling over that time period.

Exhibit 6 plots a long history of the unemployment gap and the inflation gap (spread between actual inflation and inflation expectations). The flattening of the *Phillips* curve has made forecasting inflation more uncertain and likely means that the Fed needs to see evidence in the incoming data that inflation is starting to respond to changes in the unemployment rate. Over the past year, inflationary pressures – including wage, pipeline and consumer price inflation – have been remarkably absent in the economy despite the “substantial improvements” in the labor market.

If raising rates is ultimately dependent on when inflation begins to respond to unemployment, it is appropriate for the Committee to begin to re-direct market attention toward inflation. After all, inflation is the ‘cost’ of keeping rates on hold. When the costs are low, there is little reason to raise rates. Since the Committee implemented the rate guidance framework last December, we have suggested the unemployment threshold would be the binding constraint. In other words, the unemployment rate threshold would be crossed before the 2.5% inflation threshold. With Wednesday's statement, the Committee provided some qualitative guidance that low inflation might become the binding constraint, causing the Committee to remain on hold. At the press conference, Bernanke affirmed this by saying,

But, as the last phrase of the enhanced guidance underscores, the prospects for inflation provide another reason to keep policy accommodative. The Committee is determined to avoid inflation that is too low, as well as inflation that is too high. And it anticipates keeping rates low at least until it sees inflation clearly moving back toward its 2 percent objective.

Finally, the ‘blue dots’, the mean and median of which fell, also illustrate that the Committee needs to see inflation rising before raising rates. The decline in the mean and median of the ‘blue dots’ is consistent with the downward revisions to

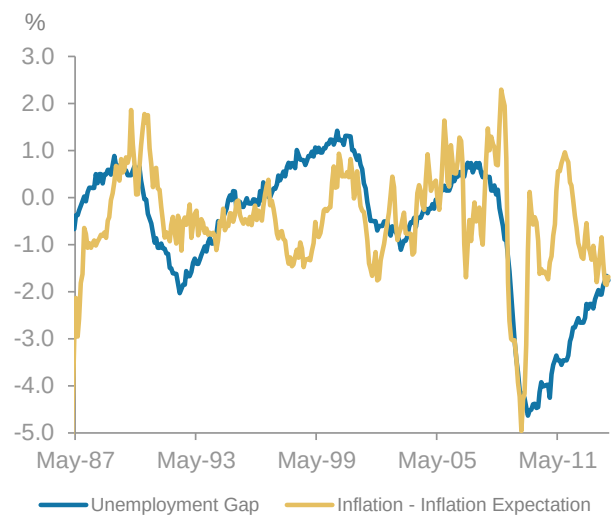
both the headline and core inflation forecasts. Moving forward, we expect the Fed's projected rate path to be more sensitive to changes in its inflation projections.

Did the Fed Change Our Market Views?

Yes. Given we believe that the chance of a cut to the URT is no longer a near-term consideration, we exit our 5s/10s and 5s/30s Treasury yield curve steepeners.

Exhibit 5

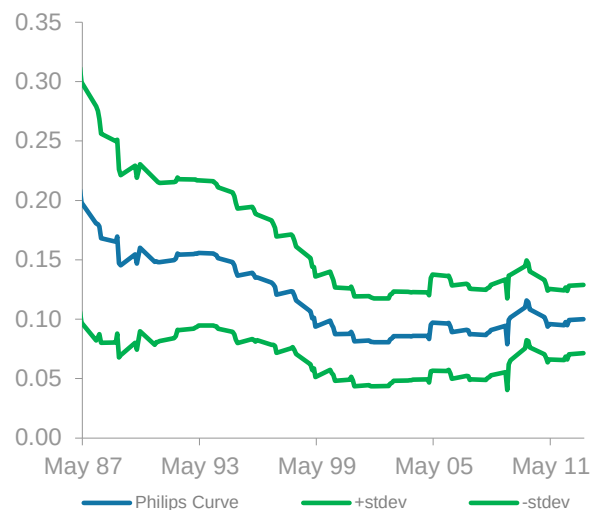
Inflation and Unemployment Gaps



Source: Morgan Stanley Research, Bloomberg

Exhibit 6

The Slope of the Phillips Curve



Source: Morgan Stanley Research, Bloomberg

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Now, we value the Treasury market in part based on what is priced in for the rate path relative to FOMC participants' median/modal expectations. On this score – as displayed in **Exhibit 1** – the market is fairly priced relative to the December SEP.

Going forward, there are two questions for markets: (1) when will the unemployment rate cross below the 6.5% threshold, and (2) how does other economic data – including realized inflation – affect the Fed's view of the appropriate pace of policy firming? In other words, how will the market define the period of time the Fed will remain on hold after the UR has crossed below the 6.5% threshold?

At present, the easiest way for the market to determine this is by looking at the 'blue dots' from the SEP relative to the unemployment rate projections. Currently, the median FOMC participant thinks the Fed should remain on hold about 9 months after the median FOMC participant (could be a different participant) thinks the unemployment rate will cross below 6.5%. In the press conference, Bernanke reminded us that the SEP represented individual assessments, not the Committee's collective view:

The SEP, the Survey of Economic Projections, which were distributed, obviously that's individual assessments and not the Committee's collective view. But nevertheless, it gives you some sense of current expectations about the length of time.

While looking at participants' assessments of policy is easy, projecting how their assessments will change over time is considerably harder. Not only does the market not know which assessment is associated with each participant, but the market also does not know the reaction function that relates each assessment with the incoming economic data. And while motivated investors could try modeling the reaction function of certain sets of policy paths based on information provided by FOMC participants in speeches or on regional Federal Reserve Bank websites, the Fed only delivers an SEP update every 12 weeks – so the data set is not very robust.

Meanwhile, the market continues to trade and investors have to interpret how the 'blue dots' will change in the interim, given economic data releases. Given the Fed's decreased focus on the UR and increased focus on other economic variables including inflation, as well as the uncertainty about the Fed's rate policy reaction function, we continue to suggest investors remain long implied volatility via the options market.

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US Treasury Inflation Protected Securities

Deflated Expectations

Tiffany Wilding (212) 761-4415

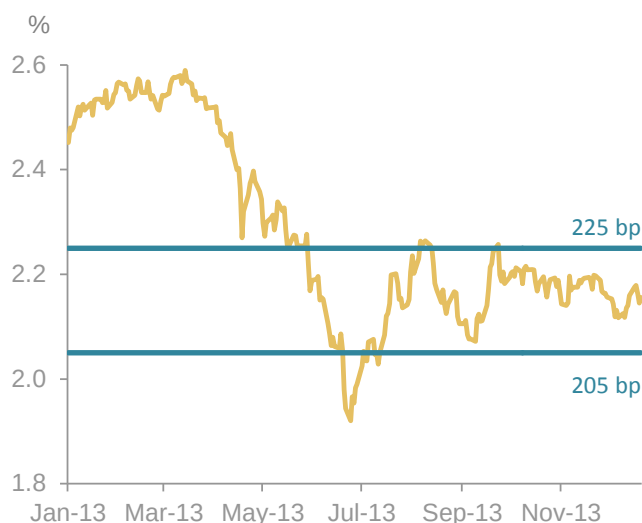
- We are turning neutral on TIPS breakevens. Based on Wednesday's FOMC statement and press conference, we believe Federal Reserve officials are not prepared to change the rate guidance framework – at least not in the “very near term.”
- Without a more accommodative change, we see little reason why TIPS breakevens should rally. The inability of the central bank to commit to a rate guidance change, coupled with our expectations that realized inflation will remain below the Fed's long-term target throughout 2014, give us reason to believe that TIPS breakevens will remain within a 225 to 205 range – at least for now.
- Since we question the credibility the market will assign to the Fed's post-6.5% unemployment ‘soft’ rate guidance, we see scope for longer-dated real rates to rally.

Turning Neutral on TIPS Breakevens

Based on Wednesday's FOMC statement and press conference, we believe Federal Reserve officials are not prepared to change the rate guidance framework – at least not in the “very near term.” Without a more accommodative change, we see little reason why TIPS breakevens should rally. The inability of the central bank to commit, coupled with our expectations that realized inflation will remain below the Fed's long-term target throughout 2014, give us reason to believe that TIPS breakevens will remain within a 225 to 205 range – at least for now (see **Exhibit 1**).

Some investors may characterize the inflation-based rate guidance included in the FOMC statement as a “soft” inflation floor. However, based on the time inconsistencies faced by central bank policy makers, it is important to emphasize that it's not a floor that will fully influence market pricing unless it's (1) included as another parameter within the current rate guidance and (2) the market believes the rate guidance framework is durable enough to divert the Fed's future behavior from the incentives that it faces.

Exhibit 1
TIPS 10-Year Breakeven Inflation Rate History



Source: Morgan Stanley Research, Federal Reserve

Only Credible Pledges Provide Accommodation

Ever since September, when the White House announced the nomination of Janet Yellen as the next chairperson of the Federal Reserve, we've been bullish on TIPS breakevens. The basis of our view was that a Yellen-led Fed would be biased to (1) provide more accommodation to bring down unemployment more quickly and (2) tolerate some above-target inflation. Furthermore, above-target inflation would allow the Fed to sustainably hike interest rates, and interest rate normalization would lessen the risks of secular stagnation – a topic that has received more attention in academic circles more recently. Our bullish view was also based on what we saw as an increasing probability the Fed would change one of the rate guidance thresholds, providing the market with a hard commitment to this strategy.

A hard commitment in the form of a rate guidance change is important, because policy makers face a time-inconsistency problem. Time-inconsistent policy describes a situation in which policy makers' preferences change over time, and, if acted upon, these policies can result in sub-optimal outcomes. In the current environment, policy makers have an incentive to commit to keeping rates low; however, when the economic situation improves – inflation and unemployment are near their respective targets – central bankers will have an

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incentive to renege on their commitments.⁵ Because forward-looking markets understand these changing incentives, the market will never price rates low enough to provide the accommodation necessary today, in our view.

Hence, to provide accommodation, policy makers must 'solve' the time-inconsistency problem by building a framework that can credibly ensure they do not renege on their commitments. In our view, this was the basis of the introduction of the rate guidance framework last December. This thinking is also the basis of our view that the soft commitments within the December FOMC statement will result in tighter financial conditions than those that would have prevailed if the Fed had changed the unemployment rate threshold or implemented the inflation floor.

At Wednesday's press conference, Bernanke signaled the FOMC is not ready to make the hard commitment necessary to achieve the optimal control-like outcomes in the paper presented in November written by William English and other authors. Instead, the FOMC official communications reflected a committee that wanted to refocus our attention toward inflation and away from the rate guidance framework and in particular the 6.5% threshold, which could be crossed by 2H2014 due to the expiration of unemployment benefits.

Why Introduce 'Soft' Inflation-Based Guidance?

Wednesday's FOMC statement also introduced "soft" inflation-based guidance on what the Fed will do when the 6.5% unemployment threshold is crossed. In particular, the statement specified that the committee intends to keep policy rates on hold "well beyond" the time the 6.5% threshold is crossed "especially if" inflation is running below the committee's 2% target. We believe the committee wanted to refocus market attention toward the inflation side of the mandate because ultimately, the committee does not know when inflation will begin to react to changes in the unemployment gap. The 6.5/2.5% unemployment/inflation thresholds were based on a guess – albeit an educated guess – about the level at which inflation would begin to react to incremental changes in the unemployment rate.

It's no secret that the Phillips curve – the relationship between unemployment and inflation – has been flattening since the 1970 US hyper-inflation. The flattening of the Phillips curve has made forecasting inflation more uncertain and likely means that before Fed officials begin to tighten policy, the

Fed needs to see evidence in the incoming data that inflation is starting to respond to changes in the unemployment rate.

Over the past year, inflationary pressures – including wage, pipeline and consumer price inflation – have been remarkably absent in the economy despite the "substantial improvements" in the labor market. Based on this, it's appropriate that the Fed would want to refocus markets toward the inflation side of the mandate. However, as we previously stated, doing this via a "soft" commitment isn't credible in our view.

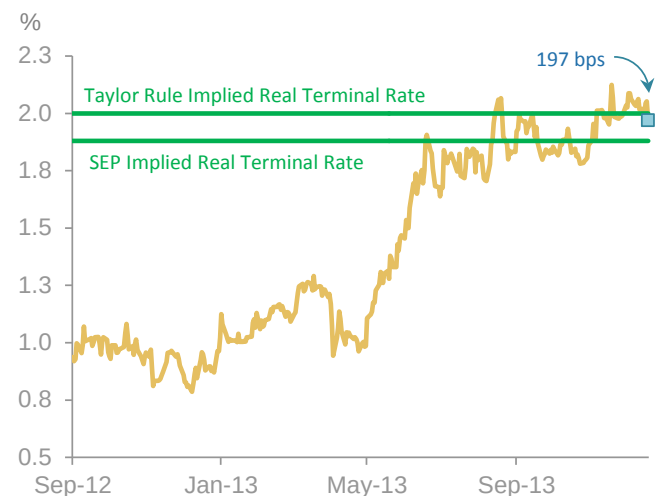
Breakevens Are Not Buying the Soft Signals

Since the Yellen nomination announcement, TIPS breakevens have traded within a very tight range. The asset class has not behaved in a way that would indicate the market perceives inflation risks as higher. Based on the price action, we believe the TIPS market does not buy into the "soft" signals that Fed officials have been sending.

Based on this, and Bernanke's press conference guidance that an announcement lowering the unemployment rate threshold should not be expected in the "very near term," we see fewer reasons why TIPS breakevens should rally. In our view, inflation markets need to see hard evidence that the committee is going to allow some above-target inflation. This, plus our economists' view that realized inflation will remain below the committee's long-run goal through 2014, has made us neutral on breakeven valuations. Hence, we are exiting our 5- and 10-year breakeven trades.

Exhibit 2

TIPS 10y20y Real Interest Rate History



Source: Morgan Stanley Research, Federal Reserve

⁵ For a simple illustration of this idea see the appendix section in the lead article on page 10.

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Long-dated Real Rates Set to Rally

In our recently published 2014 Year Ahead Outlook, we asserted that FOMC policies could drive longer-dated real forward rates higher in 2014.⁶ Wednesday's FOMC statement also changed this view.

The level of the longest-dated real forward rates should be tied to the market's rational expectations of the long-term terminal real growth rate plus term premium and a convexity adjustment. In our year-ahead report, we suggested that rational expectations for the long-term terminal real fed funds rate could rise in response to a recently released paper by senior Fed staffer, David Wilcox, asserting that aggressive post-crisis monetary policies could bring realized growth back to potential, while also bringing trend growth back toward pre-crisis levels.⁷

According to Wilcox, pre-crisis trend growth is achieved through more aggressive monetary policies. Since we question the credibility that the market will assign to the Fed's soft rate guidance, we see scope of longer-dated real rates to rally. **Exhibit 2** provides a plot of the 10y20y real rate relative to (1) the most recent SEP implied long-term terminal real rate and (2) the traditional 2% Taylor rule implied real terminal rate.

⁶ See [2014 Global Rates Outlook: Great Expectations](#), December 6, 2013

⁷ See <http://www.imf.org/external/np/res/seminars/2013/arc/pdf/wilcox.pdf>

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Short-Duration Strategy

A New Way Forward

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- We expect repo rates to trade in the 8-15bp range in the first quarter of 2014. The introduction of Treasury FRNs and the continued decline in bill supply to pressure front-end rates lower.
- The Fed announced that it will reduce the pace of asset purchases starting in January, but provided more accommodation by strengthening forward guidance and lowering projected target fed funds rate in the SEP.
- The Fed did not announce any adjustments to the overnight reverse repo facility in this meeting, but we expect the Fed to extend the program after the operational exercise ends in January.
- Traditional monetary policy that focuses on managing the supply of reserves is less relevant in today's market with large excess reserves and a large Fed balance sheet.
- As the fed funds target rate loses its efficacy, a combination of IOR and unlimited fixed rate RRP should become important policy tools.
- The suspension of the debt ceiling ends on February 7. Although we expect Congress to address the debt ceiling issue in a timely fashion, the amount of refunds will dictate whether the Treasury will be able to fund the government until mid-March when they start receiving tax payments.

More Accommodation

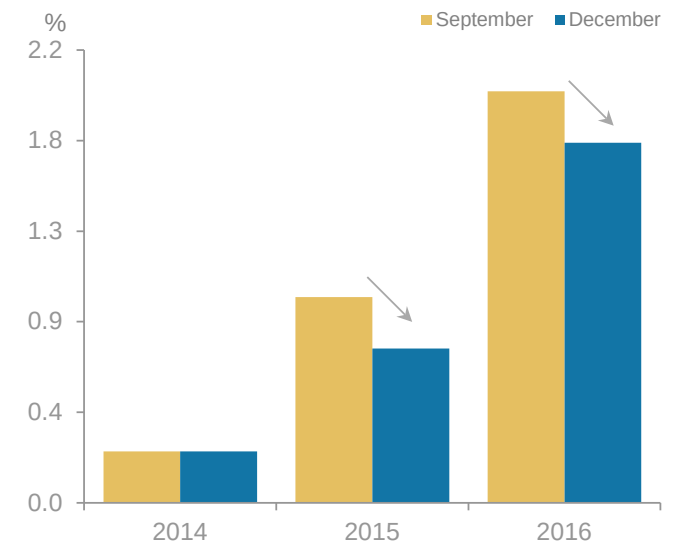
The message for front-end markets from this week's FOMC meeting is more accommodation. The Fed announced it will reduce its pace of asset purchases starting in January – by \$5bln in Treasuries and \$5bln in mortgages, to \$40bln and \$35bln respectively – but provided more accommodation by strengthening forward guidance and lowering the projected target fed funds rate in the summary of economic projections (SEP). The median projections for the fed funds target rate declined by 25bp for 2015 and 2016 (see **Exhibit 1**).

The Fed further strengthened the “lower for longer” message by stating that it expects to keep rates low “well past the time that the unemployment rate declines below 6-1/2 percent, especially if projected inflation continues to run below the committee's 2 percent long-run goal”. Essentially the Fed moved its “blue dots” to match the fed funds futures market, which was anticipating the first rate hike for November 2015. As a result, fed fund futures barely budged after the FOMC meeting on Wednesday. Since the September FOMC meeting, fed fund futures have steadily decoupled rate-hike

expectations from the timing of tapering. Markets were pricing for the first rate hike to begin in September 2014 prior to the September FOMC meeting; this has moved to November 2015 recently (see **Exhibit 2**).

Exhibit 1

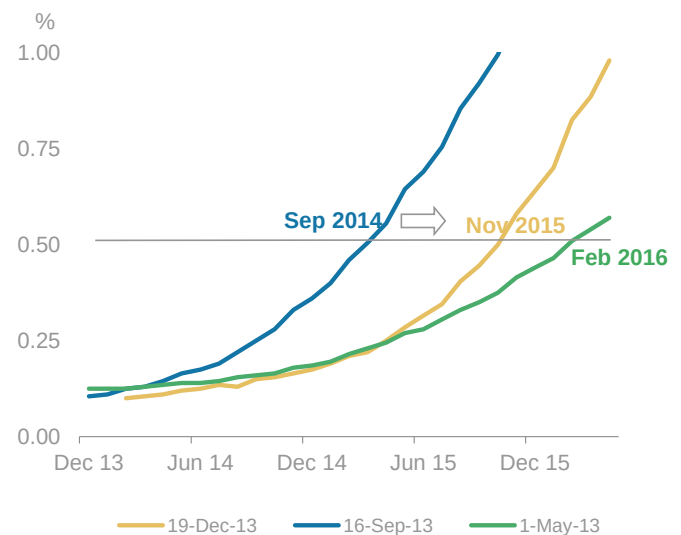
SEP Median Appropriate Pace of Policy Tightening



Source: Morgan Stanley Research, Federal Reserve

Exhibit 2

Fed Fund Futures Rate Curve



Source: Morgan Stanley Research, Bloomberg

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The Fed did not lower the interest on reserves (IOR) or make adjustments to the overnight reverse repo facility (RRP) at this week's meeting. Both rates are expected to become key policy tools of the future for the Fed. Although rate hikes are at least two years away, the current environment of abundant liquidity and large Fed balance sheet is going to warrant unconventional policy tools such as IOR and RRP. The current operational exercise for the fixed rate overnight RRP concludes on January 29, 2014. Even though the daily volumes and number of bidders were low at times, the facility has been somewhat successful in pushing front-end rates higher. Front-end bills that used to trade less than 0.5bp prior to the introduction of this facility are now trading at 1.5-2bp. We expect the Fed to extend this program at the end of operational exercise, and it will probably be available to market participants at least in its current capacity (i.e., limited size and low fixed rate) for the foreseeable future, because of its importance as a policy tool for the Fed.

A New Paradigm

The era of large excess reserves has significantly reduced the importance of the fed funds market, as banks do not need to borrow in the interbank market to meet reserve requirements. According to Fed estimates, volume in the fed funds market has declined from \$200bln in 2007 to \$60bln in 2012. Most of the lending in the fed funds market⁸ is from institutions such as the GSEs that do not receive interest on excess reserves. Hence, true interbank lending only represents a small fraction of the current fed funds market.

On the other side of the equation, it turns out that foreign banks are the largest borrowers⁹ in the fed funds market. According to the Fed, European institutions, followed by Asian and Canadian banks, are large borrowers in the fed funds market. Here again incentives are the key driver, not borrowing needs. Foreign banks not subject to the FDIC insurance are active in borrowing from the fed funds market at rates well below IOR and depositing at the Fed to earn 25bp, thus engaging in a risk-free arbitrage. Since domestic banks are subjected to FDIC insurance, the effective fed funds rate has to be much lower than IOR to compensate them for the additional insurance costs to make it worthwhile for them to participate in the arbitrage. Hence, participation from domestic banks is less than foreign bank participation. Under the current paradigm, foreign and domestic banks are key players in taking down excess reserves generated by the Fed and in the implementation of monetary policy (see **Exhibit 3**).

⁸ See [Who's Lending in the Fed Funds Market?](#) By Gara Afonso, Alex Entz, and Eric LeSueur, December 2, 2013

⁹ See [Who's Borrowing in the Fed Funds Market?](#) By Gara Afonso, Alex Entz, and Eric LeSueur, December 9, 2013

Exhibit 3

Cash Assets of Domestic and Foreign Banks



Source: Morgan Stanley Research, Bloomberg

In the current market environment with a large Fed balance sheet and large excess reserves, traditional monetary policy that is focused on managing the supply of reserves is less relevant. The future of monetary policy will hinge on the Fed's ability to effectively control the level of short-term interest rates while maintaining the desired level of accommodation. This was why IOR was introduced during the financial crisis. But for the reasons outlined above IOR has been ineffective in setting a floor on front-end rates. The new RRP facility, especially if offered in unlimited size, should prove to be effective in setting a floor on front-end rates. The RRP facility essentially offers IOR for non-bank entities such as money funds and GSEs, thus complementing IOR and strengthening the floor on front-end rates. As the fed funds target rate loses its efficacy, a combination of IOR and unlimited fixed rate RRP should become important policy tools.

Should the Fed Abandon the Target Rate?

Prior to the financial crisis, the Fed would control interest rates around its target fed funds rate by managing the amount of reserves in the system. As highlighted in a recent speech by Simon Potter¹⁰, the head of the Markets Group at the New York Fed which is charged with conducting open market operations, in the five years prior to the financial crisis the "system functioned with an average of \$20 billion in reserve balances and \$1.5bln in excess reserves". In contrast, just this week the Fed's balance crossed the \$4 trillion mark and excess reserves exceed \$2.5 trillion.

¹⁰ See [Recent Developments in Monetary Policy](#), December 2, 2013, speech by Simon Potter, New York Fed

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Although the Fed reaffirmed its target range for the fed funds rate at 0-25bp in the current statement, its usefulness as a policy tool is increasingly coming into question. If the Fed were to hike rates by increasing the fed funds target rate and by increasing IOR to, say, 50bp, it is highly unlikely that front-end rates would move higher in lockstep with it. The Fed would have to also employ the unlimited fixed-rate RRP by possibly setting it to the same level as IOR to be able to move front-end rates close to the targeted level of 50bp.

What will happen to volumes in the fed funds market?

In a recent publication¹¹, we argued that a cut in IOR will significantly impact volumes in the fed funds market where daily volumes have already declined dramatically since the financial crisis. Similarly if the Fed were to set the fixed rate on the unlimited RRP equal to IOR, then the arbitrage that currently exists in the fed funds market will all but disappear and volumes are bound to decline. There will be some small banks that might continue to use the interbank market to meet reserve requirements, but most banks will not need to participate in the fed funds market. GSEs will instead redeploy their cash into the Fed's RRP, where they can earn a risk-less return without any credit exposure.

What about instruments linked to the fed funds rate?

Before the Fed decides to cut IOR or to set IOR and RRP to the same rate level, serious consideration should be given to the impact not just on the fed funds market but also on instruments such as floating rate notes (FRN) and overnight index swaps (OIS) that are linked to the fed funds effective rate. One approach would be to create an alternative index that better represents a rate derived from market transactions. In a recent proposal released for comments in June, the Fed would require certain depository institutions to report a selection of money market rates based on daily transactions in fed funds, Eurodollars and CDs. This data could be combined to create a "broad-based unsecured dollar rate" that could be used as a replacement for interbank rates such as fed funds effective and LIBOR.

Market Outlook

We expect repo rates to trade in the 8-15bp range during the first quarter of 2014.

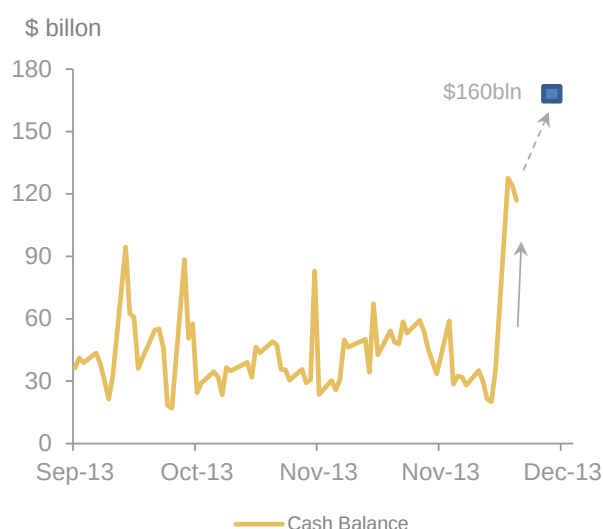
Bill Supply

Over the near term, the reduction in bill supply should continue to keep front-end rates low. The Treasury has reduced the 4-week bill issuance sizes for the past few weeks and we could see further reductions in January as the

Treasury's cash position continues to improve. The GSEs (Fannie Mae and Freddie Mac) combined are expected to make another large dividend payment of \$39bln on December 31. The Treasury currently has \$117bln in cash on hold at the Fed. After the large dividend payment, our economists estimate that the Treasury will have \$160bln in cash to start the year (see **Exhibit 4**). We expect net bill supply to increase only modestly in the first quarter, owing to the Treasury FRN issuance and the Treasury's cash position.

Exhibit 4

Treasury Cash Balance



Source: Morgan Stanley Research, Bloomberg

Debt Ceiling

The suspension of the debt ceiling expires on February 7, after which the Treasury will need to start employing extraordinary accounting measures to continue to fund the government. The Treasury secretary, Jack Lew, sent a letter to Congress this week reminding them of the looming deadline and that the Treasury will not be able finance the government beyond late February or early March. Although we expect Congress to address the debt ceiling issue in a timely fashion in January, the amount of tax refunds will dictate if the Treasury can make it to March when tax payments could tide them over through mid-year. Due to the government shutdown, the IRS will begin processing tax filings beginning January 31 as opposed to the usual start date of January 21. On margin, this is negative as refunds after the February 7 deadline will further strain the Treasury's cash management efforts. If the debt ceiling is not raised in a timely fashion, this could again put pressure on bill supply and push front-end rates lower.

¹¹ See [US Interest Rate Strategy Insight – Pulling on a String](#), November 22, 2013

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US Interest Rates Derivatives

Changing Guidance Means Cheap Belly: Enter Payor Flies

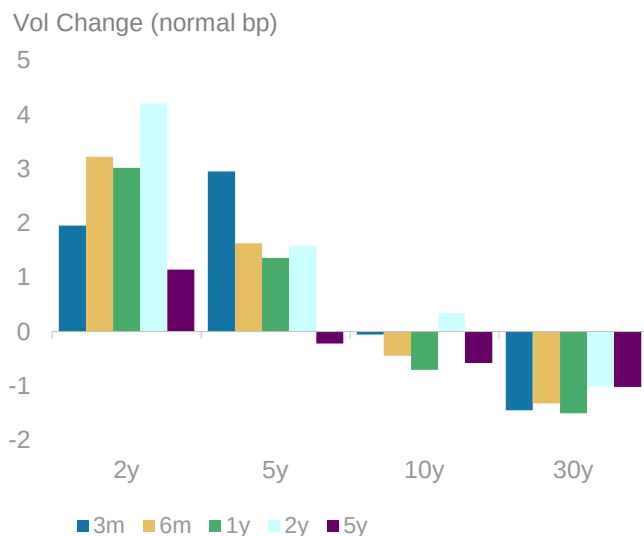
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- Vols rose this week, led by the upper-left of the surface as a natural weakening in FOMC forward guidance drove a bear-flattening of the swap curve. We believe the move is justified, and opine on what the change in tone may imply for vol.
- A 4s/7s/10s payor butterfly hedges a continued underperformance in belly rates. We select the fly as it has optimal risk-adjusted rolldown and carry, and offers better value than a simple long payor position.

This week's FOMC statement triggered a large sell-off on Thursday, led by the 4-year point, as markets processed the looming expiry of unemployment-threshold-based forward rate guidance. The price action, which drove 5-year rates 7bp higher close-to-close and even more intraday, pushed the upper-left of the vol surface higher (see **Exhibit 1**). In our view, the subsequent price action in both rates and vol is justified, and the trend may continue in the near term.

Exhibit 1

Change in Normalized Vol*



* Changes calculated from December 13, 2013 to December 20, 2013.
Source: Morgan Stanley Research, Bloomberg

FOMC Takeaways

The FOMC statement and press conference have changed the way markets should interpret economic data. Chairman Bernanke stated in his press conference that

In determining how long to maintain a highly accommodative stance of monetary policy, the Committee will also consider other information, including additional measures of labor market conditions, indicators of inflation pressures and inflation expectations, and readings on financial developments.

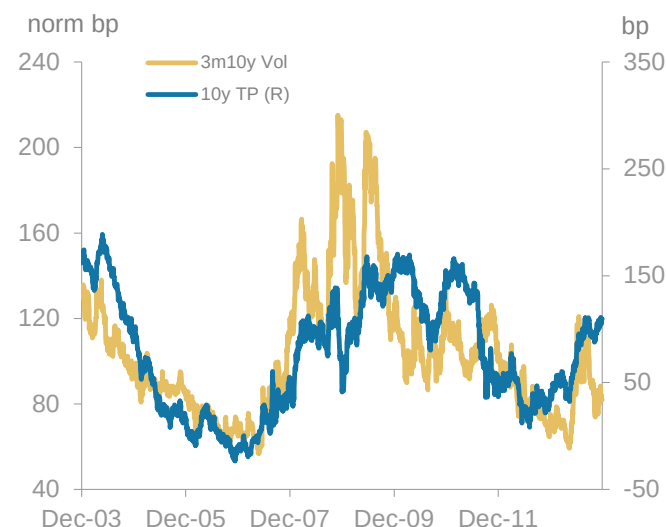
In our view, the change in tone implies that, once the unemployment rate falls below 6.5%, the FOMC's credible unemployment threshold framework for forward rate guidance will revert to a vaguer framework. As a result, **we view the new language as allowing forward rate guidance to naturally weaken**. We believe that the change in FOMC guidance has important consequences for realized and, correspondingly, implied vol:

- **Vol should rise.** The timing of liftoff can no longer be simply measured by the level of unemployment. Markets will need to make educated guesses about the Fed's reaction function, and only receive feedback via Fed communications – speeches, statements, Summary of Economic Projections (SEP), etc. We analyzed the impact of the Evans' Rule framework on volatility in the past, and found that the framework helped to distribute realized vol over data, with a bias towards lowering vol¹². The heightened uncertainty over Fed policy should lead to higher vol, **especially around Fed communications** when markets will be able to 'mark-to-market' its expectations.
- **Higher belly term premiums.** A rise in uncertainty and implied vol should help push term premiums higher, centered in the belly of the yield curve where rates are most sensitive to the timing of liftoff. However, we acknowledge that relative to volatility, term premiums are relatively cheap (see **Exhibit 2**). As such, we prefer going long vol over outright short-duration positions.
- **Skews should normalize.** Yesterday's statement shifted language on the balance of economic risks, from focused on downside risks to balanced. Moreover, the advent of tapering provides greater flexibility to the Fed to increase accommodation (by either maintaining or increasing asset purchases), whereas the hurdle to increasing purchases was higher previously. As a result, **potential surprises in policy are also more balanced**. We believe the disconnect between market pricing and

¹² See [US Interest Rate Strategist: High Enough?](#), May 3, 2013.

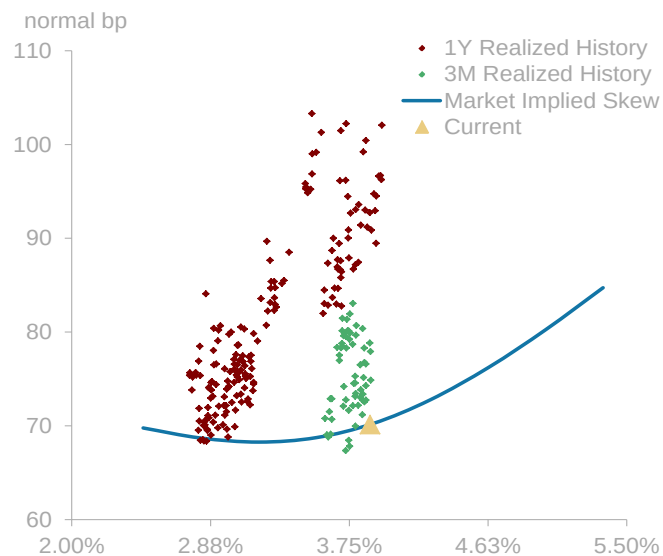
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Exhibit 2
3m10y Implied Vol vs. 10-Year Term Premiums*



* Term premiums are calculated using our Dynamic Term Structure Model; please consult your MS representative for more information.
Source: Morgan Stanley Research

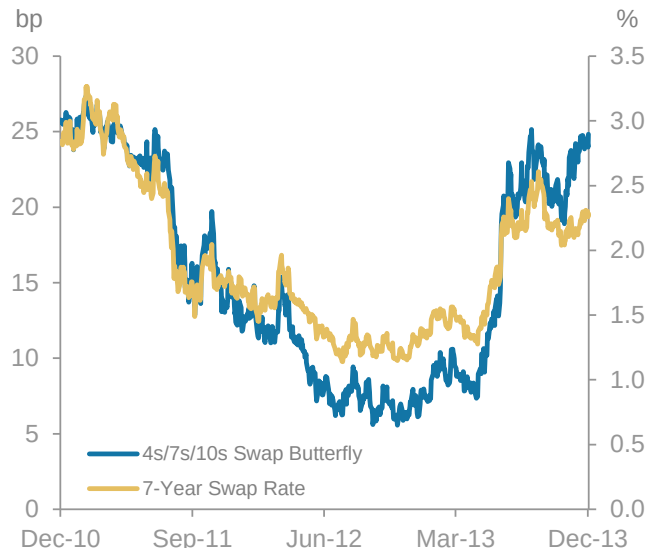
Exhibit 3
3m10y Forward Swap Rate and Implied Vol vs. Market Implied Skew



Source: Morgan Stanley Research

implied skew levels is most severe in 30-year tails. **Exhibit 3** shows that 3m30y implied vol over the past year has had no consistent relationship with the level of rates. The blue line shows that, despite the weak correlation between rates and vol, markets continue to imply high-strike vol at a premium relative to ATMF

Exhibit 4
4s/7s/10s Butterfly vs. 7-Year Swap Rate



Source: Morgan Stanley Research

options. Moreover, our prior analysis indicates that long-dated forward rates were approaching the ceiling of what the Fed's stated longer-run short rate would imply after adjusting for convexity and term premium. The slight decline in the December SEP longer-run short rate, from 3.93% to 3.88%, adds fuel to the idea that long-dated risks should be more balanced.

We don't expect the FOMC statement to have a meaningful impact on swap spreads. The advent of tapering reduces our expectations for net Treasury purchases over 2014 by \$100bn (see this week's Governments section), pushing swap spreads tighter. On the other hand, the weaker forward rate guidance may cause rate expectations to rise, which is well correlated with wider swap spreads. Net, we continue to see 10-year spreads as 6-7bp too wide at current levels.

Trade Idea: Enter 3m 4s/7s/10s Payor Flies

As discussed earlier, we expect the change in Fed policy to push belly rates and vols higher. We recommend positioning for such a move via the following trade:

- Buy \$100mm 3m7y 5bp OTM payors (2.48% strike)
- Sell \$84mm 3m4y ATMF payors (1.31% strike)
- Sell \$37mm 3m10y ATMF payors (3.05% strike)
- Trade is zero cost and 50:50 PV01 weighted
- Hold to maturity

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Exhibit 4 graphs the 4s/7s/10s fly over the past three years and against the 7-year swap rate. The two series are well correlated, with the fly level currently appearing cheap to the level of rates. We believe that the cheapening of the belly can continue, driven by a bear-flattening of the curve as the Fed reduces accommodation (see the US Governments section for more information on our macro rationale).

We select the 4s/7s/10s fly by screening across various swap flies and filtering for the best risk-adjusted rolldown and carry (RDC). **Exhibit 5** tables a list of fly combinations where the short-belly position is in the 5-7 year sector.

Since the belly is relatively steep, most short-belly flies have net punitive risk-adjusted RDC. However, paying RDC to enter certain flies is still attractive relative to buying outright swaption payors. We can write the formula for an ATMF swaption payor, quoted in bp running, as:

$$\sqrt{\frac{1}{2\pi}} \times \sigma_{normal} \times \sqrt{t}$$

If we risk-adjust assuming that realized volatility is equal to implied volatility over the trade horizon, we are left with the constant term $\sqrt{\frac{1}{2\pi}}$, or approximately 0.4. If we think of the

RDC on a swap fly as the 'premium' we pay to enter the trade, then as a general rule of thumb, a fly offers better risk-adjusted performance whenever the punitive RDC 'premium' is less than 40% of the horizon-scaled realized volatility. The highlighted swap flies in the above table pick out swaps with a sufficiently low risk-adjusted performance. Ignoring flies that are highly bunched together (i.e., 6s/7s/8s) points to the 4s/7s/10s fly as not only optimal, but a superior alternative over purchasing 7-year gamma outright.

Given the high degree of correlation between fly and rate, we structure the trade as a conditional payor fly to remove exposure to a deterioration in economic data. Since the 3m7y vol is richer than the blended 3m10y and 3m4y vols, the 3m7y leg is struck approximately 5bp OTM to make the trade zero cost.

The trade profits if rates rise led by the 7-year point. If rates generally fall, the payors expire OTM, and the trade breaks even. The primary risk to the trade is that 7s outperform in a sell-off. This could occur if economic data come in much more strongly than anticipated, and the Fed's anchor on short-end rates weakens, allowing the front end to become unhinged.

SDR Data Update

Swap and swaption data continues to flow into the DTCC's swap data repository. Here is this week's data:

Exhibit 5

Fly Screener Statistics

Fly	3m RDC*	Realized Vol**	Risk Adj. RDC
1s/5s/10s	10.1	16.2	0.63
1s/7s/15s	9.4	19.2	0.49
2s/5s/7s	5.8	11.5	0.50
2s/5s/10s	7.4	13.2	0.56
2s/5s/30s	10.7	17.9	0.60
2s/7s/10s	5.0	14.7	0.34
3s/5s/7s	2.7	6.4	0.42
3s/5s/10s	4.3	8.5	0.51
3s/7s/10s	1.9	9.8	0.19
4s/5s/6s	0.4	2.1	0.20
4s/7s/10s	0.4	5.4	0.07
5s/6s/7s	0.3	1.0	0.26
5s/7s/10s	0.4	2.4	0.18
6s/7s/8s	-0.1	0.8	-0.09
7s/8s/9s	-0.1	0.7	-0.11

* Roll-down and carry (RDC) is shown in bp running on a long belly position in the fly over a 3 month horizon.

** Realized vol calculated over prior 3-months and scaled to 3-month horizon.

Highlighted rows indicate flies where risk-adjusted RDC is less than 0.4.

Source: Morgan Stanley Research

Exhibit 6

Weekly SDR Swap Volume*

Notional \$bn	1Y	2Y	3Y	5Y	7Y	10Y	20Y	30Y	Total
Spot	13.1	39.9	52.3	130.7	41.0	67.8	8.6	12.7	366.1
3m	0.2	10.1	1.4	17.7	4.3	6.6	0.9	1.2	42.4
6m	0.3	1.1	0.5	1.4	1.1	2.5	0.3	0.7	7.8
1y	7.1	4.1	0.1	4.7	0.0	1.6	0.0	1.0	18.6
2y	36.8	3.9	0.4	0.9	0.3	0.8	0.0	0.2	43.4
3y	48.2	2.1	0.1	0.2	0.0	0.3	0.1	0.0	51.0
5y	14.6	1.7	1.2	4.1	0.0	0.5	0.0	0.2	22.2
10y	0.9	2.9	0.2	0.8	0.0	0.5	0.1	0.0	5.3
Total Non-Spot	108.1	26.0	3.9	29.7	5.8	12.8	1.3	3.3	190.8
Total	121.2	65.9	56.1	160.4	46.8	80.6	9.9	16.1	557.0

*Volumes include USD 3m Libor interest rate swaps traded between December 13, 2013 and December 19, 2013 and reported to the DTCC's SDR.

Source: Morgan Stanley Research, DTCC

Exhibit 7

Weekly SDR Swaption Volume*

Notional \$mm	1Y	2Y	3Y	5Y	7Y	10Y	30Y	Total
1m	0	4,600	480	7,953	461	4,310	534	18,338
3m	0	2,980	2,420	11,737	2,470	5,355	880	25,842
6m	1,560	4,560	1,052	1,440	500	4,603	113	13,828
1y	11,200	8,080	900	5,807	300	9,548	756	36,591
2y	950	2,160	480	2,530	930	1,661	1,217	9,928
3y	6,550	1,410	634	400	0	2,480	464	11,938
5y	1,200	1,170	0	340	0	1,280	1,173	5,163
10y	0	0	0	2	0	494	220	716
Total	21,460	24,960	5,966	30,209	4,661	29,731	5,357	122,343

*Volumes include USD European swaptions traded between December 13, 2013 and December 19, 2013.

Source: Morgan Stanley Research, DTCC

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Appendix

In [US Interest Rate Strategist: A New Hope](#), March 8, 2013, we introduced our trade optimizer, which takes outright duration trades in swaps and finds alternative trades based on

RDC and relative value factors. The trades the optimizer selects are presented below, with the ones we find most attractive highlighted in gray.

Trade Selection Results

Trade	Time	Correl	Beta	Excess 3m RDC (\$K)	Regression Deviation (\$K)	Total Pickup (SD)
2-Year Pay Fixed						
ED2/ED3 Curve Steepener	1y	0.95	4.1	802	-25	3.7
	3m	0.91	3.5	802	9	5.4
ED2/ED4 Curve Steepener	1y	0.97	1.8	621	-53	3.4
	3m	0.93	1.5	654	34	5.3
10Y Forward 5y/10y Swap Curve Flatteners	1y	-0.81	-1.8	726	-104	1.6
	3m	0.18	0.6	826	214	3.0
	1y	0.71	0.8	854	-111	1.6
30Y Spread Widener	3m	0.79	1.0	870	141	4.6
10Y Forward 2y/10y Swap Curve Flatteners	1y	-0.81	-1.0	651	-191	1.2
	3m	0.33	0.7	907	308	3.6
2-Year Receive Fixed						
	1y	0.96	1.1	236	431	3.4
Long 2Y UST	3m	0.90	1.2	337	265	3.8
	1y	0.94	0.3	-104	641	2.3
Long ED8	3m	0.91	0.3	-175	302	0.8
	1y	0.96	0.4	-86	500	2.1
Long ED7	3m	0.94	0.3	-218	209	-0.1
	1y	0.91	0.2	-183	716	1.9
Long ED9	3m	0.87	0.2	-191	351	0.9
ED4/ED8 Curve Flatteners	1y	0.88	0.3	-172	722	1.7
	3m	0.79	0.4	-112	357	1.1
5-Year Pay Fixed						
ED2/ED3 Curve Steepener	1y	0.81	17.0	1831	3394	2.7
	3m	0.37	2.7	1831	1388	5.2
	1y	0.81	4.5	2129	2750	2.5
30Y Spread Widener	3m	0.21	0.5	1864	1493	5.1
ED2/ED4 Curve Steepener	1y	0.85	7.7	1057	3263	2.4
	3m	0.40	1.2	1713	1404	5.1
10Y Forward 5y/10y Swap Curve Flatteners	1y	-0.82	-9.0	1452	2913	2.3
	3m	-0.58	-3.5	1683	1113	5.1
10Y Forward 2y/10y Swap Curve Flatteners	1y	-0.82	-4.8	1074	2472	1.9
	3m	-0.49	-1.9	1537	1063	4.5
5-Year Receive Fixed						
ED6/ED10 Curve Flatteners	1y	0.99	1.3	400	1389	3.7
	3m	0.90	0.6	-720	-120	-2.9
ED4/ED8 Curve Flatteners	1y	0.97	1.9	1589	527	2.7
	3m	0.76	0.7	-583	-577	-2.7
	1y	0.99	1.0	-13	898	2.7
Long 5Y UST	3m	0.89	0.8	-309	247	-0.2
	1y	0.98	2.1	1524	-142	2.4
Long 3Y UST	3m	0.90	1.4	423	-353	0.2
Long 5y on 2y/5y/7y Swap Fly	1y	1.00	3.2	280	267	2.3
	3m	0.97	2.6	-143	-99	-1.5
10-Year Pay Fixed						
	1y	0.80	5.7	1579	4183	2.2
30Y Spread Widener	3m	-0.39	-1.2	1124	1808	3.8
ED2/ED3 Curve Steepener	1y	0.73	19.9	1202	5043	2.1
	3m	-0.31	-2.7	1202	1908	3.9
Pay 1y Forward 10y Swap	1y	1.00	0.9	148	-20	2.1
	3m	1.00	0.9	170	70	4.7
10Y Forward 5y/10y Swap Curve Flatteners	1y	-0.78	-11.1	737	4427	1.9
	3m	-0.80	-6.0	949	1135	4.2
ED12/ED16 Curve Steepener	1y	0.96	2.3	2456	-227	1.9
	3m	0.64	0.8	1633	1378	4.7
10-Year Receive Fixed						
	1y	0.99	1.0	517	1048	3.5
Long 7Y UST	3m	0.98	0.8	162	333	2.9
	1y	0.99	1.3	1148	559	3.3
Long 5Y UST	3m	0.92	1.1	752	445	3.7
ED6/ED10 Curve Flatteners	1y	0.98	1.6	1652	1124	3.0
	3m	0.52	0.5	-403	-802	-1.7
ED8/ED10 Curve Flatteners	1y	0.98	2.7	720	1133	2.4
	3m	0.60	1.0	-530	-667	-1.8
3M Forward 2y/5y Swap Curve Flatteners	1y	1.00	1.5	231	433	2.4
	3m	0.98	1.4	106	107	1.4
30-Year Pay Fixed						
	1y	0.77	4.9	845	3924	2.0
30Y Spread Widener	3m	-0.52	-1.4	426	1359	2.8
ED12/ED16 Curve Steepener	1y	0.98	2.0	1646	-88	1.9
	3m	0.80	0.9	1007	873	4.2
2Y Fwd 2y/7y Swap Curve Steepener	1y	0.81	2.4	3700	405	1.9
	3m	0.67	0.5	1258	1233	4.5
Pay 10y Forward 5y Swap	1y	1.00	0.9	546	-64	1.8
	3m	0.98	0.7	541	222	5.6
2Y Fwd 2y/5y Swap Curve Steepener	1y	0.90	2.8	3117	-233	1.7
	3m	0.72	0.7	1204	1106	4.4
30-Year Receive Fixed						
	1y	0.99	0.9	992	703	3.0
Long 7Y UST	3m	0.93	0.7	660	496	4.3
	1y	0.99	0.9	678	499	2.7
Long 10Y UST	3m	0.96	0.7	431	171	3.0
ED6/ED10 Curve Flatteners	1y	0.95	1.4	1943	659	2.2
	3m	0.27	0.2	-140	-872	-1.4
	1y	0.98	1.1	1524	214	2.1
Long 5Y UST	3m	0.77	0.8	971	366	2.8
ED8/ED10 Curve Flatteners	1y	0.97	2.4	1149	703	1.9
	3m	0.36	0.5	-157	-735	-1.3

These trades do not represent our recommendations. Please see our trade summary elsewhere in this report. *Treasury and Spread analysis based on CMT rates. RDC and regression deviation based on 100K of PV01 for the equivalent swap curve trade. Total pick-up is basis points of RDC+ regression deviation, all in terms of the standard deviation of the regression. Source: Morgan Stanley Research

US Agency Mortgage Backed Securities

Tapering Has Started – Now What?

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- While the tapering announcement has lowered our base case expectation for Fed purchases, the Fed is still expected to buy more than the 2014 net issuance projection. MBS performance is less likely to be driven by pure OAS widening alone next year. We are most concerned about bear-flattening of the 5-10y portion of the curve, an increase in volatility, and swap spread widening.
- We recommend transitioning from our 4/3.5 swap to a long position in the Fannie 4 fly. This fly has cheapened significantly recently. We are neutral on the mortgage basis and 15s/30s.

This week, the Fed announced they will reduce their purchases of agency MBS and Treasuries by \$5bn each starting in January. Although in our base case, we had expected tapering to begin next year, the odds for this scenario occurring were still relatively high. The Fed did pair the tapering announcement with stronger forward guidance, saying they plan to keep the fed funds rates low “well past” after the unemployment rate threshold is crossed.

Our rates strategists argued in their [note](#) Wednesday night that the yield curve should bear-flatten, led by the 5-year maturity point. They continue to believe that volatility should head higher as well next year. Based on the market reaction after the announcement, it seems the markets did not consider forward guidance strong enough as the curve bear-flattened and the belly of the yield curve underperformed.

For the MBS market, there are a number of implications. Due to the sooner than expected tapering announcement, the Fed, in our new base case, will not buy as much in MBS as we originally envisioned. We now expect the Fed to buy \$220bn in MBS on a net basis instead of the \$350bn we had penciled in. These new expectations are based on a \$5bn cut in MBS purchases in every meeting. Our net issuance forecast for 2014 is \$152bn.

Therefore, we believe the Fed is still going to buy more than the projected net issuance for next year and most investors will still be net sellers to the Fed. The Fed purchases are more front loaded and during the second half of the year, most of the gross supply will have to be absorbed by private market participants. Themes that we had highlighted from our 2014 Outlook are still very much intact.

- The big risk event associated with the first tapering announcement is now behind us. The Fed has indicated that further cuts will be very gradual, which means that risks related to tapering are not as binary and they can be priced in the market much more efficiently.
- As we had argued in our Outlook, if the tapering expectations are properly priced in, there should not be a major re-pricing after each announcement. Relatively speaking, the MBS market has been fairly stable despite a slightly sooner than expected tapering announcement, which underscores defensive positioning in the MBS market.
- MBS performance is less likely to be driven by pure OAS widening alone next year. We are most concerned about bear-flattening of the 5s-10s portions of the curve, an increase in volatility, and swap spread widening. These risks are much more hedgeable, but they reduce the carry profile.

Exhibit 1

Projected Supply/Demand for 2014 (\$bn)

	2010	2011	2012	2013 (YTD)	2014 (Est)	As of
Net Issuance	-147	24	30	225	152	11/30/2013
Demand						
GSEs	-260	-91	-85	-64	-40	10/31/2013
Treasury	-47	-118	-26	0	0	3/31/2012
Overseas	-20	-24	-47	-68	-20	10/31/2013
Fed	-71	-119	169	499	220	12/18/2013
Banks	112	143	88	-17	-20	12/11/2013
REITs	33	106	123	-67	0	9/30/2013
Money Managers/ Others	106	127	-192	-59	12	Residual
Net Demand	-147	24	30	225	152	

Source: Morgan Stanley Research, TICS

December 20, 2013
US Interest Rate Strategist

In terms of market positioning, we remain neutral on the mortgage basis. We continue to believe that long mortgage basis positions should be hedged more with 7y swaps while short mortgage positions should be hedged more with 10y Treasuries. We would also warn investors not to underestimate demand from other private market participants once the Fed completes the QE3 program, especially overseas investors.

We had recommended going long FN 4/3.5 on 11/14/2013. The trade is currently flat. We recommend converting the trade into long FN 4 Fly trade. FN 4 fly has cheapened a lot recently. The risk to the trade is continued outperformance in 4.5s as the liquidity in the coupon is worse than 3.5s and 4s. We had also recommended being overweight 15s/30s on 11/22/2013. Post the tapering announcement, we return to neutral.

December 20, 2013
US Interest Rate Strategist

Market Data Summary

US Treasuries

Maturity	YTM (%)	1w Chg (bp)
2y	0.366	3.8
3y	0.698	2.0
5y	1.643	11.2
7y	2.327	8.7
10y	2.923	4.6
30y	3.898	0.2
2s/3s	(bp) 33.2	-1.8
3s/5s	94.5	9.2
5s/7s	68.4	-2.5
7s/10s	59.6	-4.1
10s/30s	97.6	-4.4
2s/3s/5s	-30.7	-5.5
2s/5s/10s	-0.1	7.0
5s/7s/10s	4.4	0.8
5s/10s/30s	15.2	-1.1

US Agencies

Maturity	Spread to UST (bp)	1w Chg (bp)
2y FHLMC	-17.3	-4.3
3y FNMA	-7.0	1.0
3y FHLMC	-1.5	-1.0
3y FHLB	-23.5	-3.8
5y FNMA	12.0	-5.3
5y FHLMC	-19.5	-6.0
5y FHLB	-47.0	-8.2
7y FHLMC	0.5	-6.3
10y FHLMC	1.0	-1.2

Forward Swap Rates

Maturity	Rate (%)	1w Chg (bp)
1y x 1y	0.59	2.11
1y x 2y	1.02	2.25
2y x 2y	2.04	5.65
2y x 5y	3.19	6.76
3y x 2y	3.08	9.53
3y x 5y	3.83	7.49
3y x 10y	4.30	6.04
4y x 5y	4.28	6.81
5y x 5y	4.55	5.15
5y x 10y	4.72	4.91
10y x 10y	4.83	5.33
10y x 20y	4.66	4.13

Treasury GCF Futures

Contract	Rate (%)	1w Chg (bp)
GCF Jan14	0.075	0.0
GCF Feb14	0.105	0.0
GCF Mar14	0.105	0.0
GCF Apr14	0.120	-0.5
GCF May14	0.130	-0.5
GCF Jun14	0.135	-0.5

Eurodollar Futures

Pack	Rate (%)	1w Chg (bp)
Whites	0.34	5.6
Reds	0.72	16.1
Greens	1.74	29.6
Blues	2.89	32.0

USD Swaps

Maturity	YTM (%)	1w Chg (bp)
2y	0.444	2.0
3y	0.782	2.1
5y	1.673	4.9
7y	2.361	5.2
10y	2.988	4.9
30y	3.846	4.2
2s/3s	(bp) 33.7	0.1
3s/5s	89.1	2.8
5s/7s	68.8	0.3
7s/10s	62.7	-0.3
10s/30s	85.8	-0.7
2s/3s/5s	-27.7	-1.3
2s/5s/10s	-4.3	1.5
5s/7s/10s	3.0	0.3
5s/10s/30s	22.8	0.3

Mortgage Pass-Through OAS

Coupon	LIBOR OAS	1w Chg (bp)
FNMA 3.5	35	0
FNMA 4.0	30	2
FNMA 4.5	22	2
FNMA 5.0	22	-2

Mortgage Performance vs. Swaps

Coupon	Price	1w Perf
FNMA 3.5	99-12+	-0-070
FNMA 4.0	102-31+	-0-090
FNMA 4.5	106-01+	-0-050
FNMA 5.0	108-19+	0-01+

Swaption Implied Volatility

Maturity	Norm Vol (bp)	1w Chg (bp)
3m x 2y	33.3	2.6
3m x 5y	77.4	-0.7
3m x 10y	80.4	-3.6
6m x 2y	41.9	1.5
6m x 5y	84.7	0.0
6m x 10y	86.0	-2.8
1y x 10y	92.6	-2.0
2y x 5y	104.4	0.2
3y x 10y	99.2	-0.7
5y x 5y	104.7	-0.2
5y x 10y	96.9	-0.4
10y x 10y	81.6	-0.3

Fed Monitor

Meetings	Expected Rate (%)	1w Chg (bp)
Target	0.125	
12/18/2013	0.08	-0.3
1/29/2014	0.08	-0.5
3/19/2014	0.09	-0.3
4/30/2014	0.09	0.0
6/18/2014	0.11	0.8
7/30/2014	0.11	1.5
9/17/2014	0.11	1.0
10/29/2014	0.11	1.0
12/17/2014	0.12	1.8
1/28/2015	0.14	2.0
3/18/2015	0.16	2.8
4/29/2015	0.19	3.5

UST Benchmark Swap Spread

Maturity	Spread (bp)	1w Chg (bp)
2y	7.8	-1.8
3y	8.4	0.1
5y	3.0	-6.3
7y	3.4	-3.5
10y	6.5	0.2
30y	-5.3	4.0
2s/3s	0.5	1.9
3s/5s	-5.4	-6.3
5s/7s	0.4	2.8
7s/10s	3.1	3.7
10s/30s	-11.8	3.8
2s/3s/5s	3.0	4.1
2s/5s/10s	-4.2	-5.5
5s/7s/10s	-1.4	-0.5
5s/10s/30s	7.6	1.4

Ginnie/Fannie Swaps

Swap	Spread (32nds)	1w Chg
GN/FN 3.5	1-130	0-050
GN/FN 4.0	1-010	0-03+
GN/FN 4.5	0-230	0-02+
GN/FN 5.0	-0-060	-0-01+

Gold/Fannie Swaps

Swap	Spread (32nds)	1w Chg
FG/FN 3.5	-0-05+	0-010
FG/FN 4.0	-0-07+	0-00+
FG/FN 4.5	-0-04+	0-01+
FG/FN 5.0	-0-200	0-000

US TIPS - Real Yields

Maturity	Real Yield (%)	1w Chg (bp)
TI17/15	-1.48%	-6.7
TI14/18	-0.36%	5.3
TI17/23	0.75%	4.3
TI12/43	1.62%	0.5

US TIPS - Breakeven Inflation

Maturity	BEI (bp)	1w Chg (bp)
TI17/15	173.92	5.8
TI14/18	170.80	0.7
TI17/23	200.75	0.3
TI12/43	229.53	-0.4

Other Market Data

Index/Security	Close	1w Chg
MS FCI	-0.60	-0.12
S&P 500	1,809.60	34.10
DJ Industrials	16,179.08	439.65
USD/JPY	104.240	0.880
USD/EUR	1.366	-0.009
10y Bund	1.87%	2.86
10y JGB	0.68%	1.99
10y Gilt	2.96%	5.53
Gold	1189.90	-36.00
Gasoline	274.01	10.53
3m Libor -OIS	14.88	-0.90
CDX NA IG	66.12	-4.13
VIX	14.15	-1.4

Source: Morgan Stanley Research
All levels as of close: 19-Dec-13

Current Trade Ideas

Trade	Rationale	Entry Date	Close Date	Levels					Current chg
				Entry	Current	Objective	Re-assess	Units	
Treasuries									
Short 5s 2s/5s/30s (50:50 Weighted)	The rate path priced by the market will require a higher level of term premium	12/20/2013	-	-97.82	-97.82	-80.00	-115.00	bp running	0.00
Short 7s 3s/7s/10s (50:50 Weighted)	The rate path priced by the market will require a higher level of term premium	12/20/2013	-	109.30	109.30	125.00	95.00	bp running	0.00
3s/5s v 10s/30s Flatteners (1.0:0.5 DV01 weights of curves)	The rate path priced by the market will require a higher level of term premium. This trade allows for a less directional play that retains the potential to profit from lower rate guidance credibility	12/20/2013	-	91.46	91.46	75.00	105.00	bp running	0.00
Swaptions									
1x2 3m5y5y Payors 4.218/4.488	Based on our fundamemntal views for rates and the economy, we think it would be difficult for the 5y5y forward rate to selloff beyond 4.76%.	10/17/2013	-	0.00	5.00	15.00	-10.00	bp running	5.00
Buy 3m7y 5bp OTM payor, sell 50/50 weigh 3m4y and 3m10y ATMF payors	As forward guidance weakens we expect the belly of the curve to underperform in rates, and that vol to outperform.	12/19/2013	-	0.00	0.00	10.00	-10.00	bp running	0.00
Mortgages									
Long FN 4 Fly	The FN 4 fly has cheapened significantly recently. We recommend transitioning out of our 4/3.5 swap into the fly as a more attractive position.	12/20/2013	-	17.00	17.00	22.00	12.00	ticks	0.00

Note: The portfolio represents hypothetical, not actual, investments. Profit and loss does not include expenses such as commissions and fees. Past performance is no guarantee of future results.

Please see [US Interest Rate Strategy - Historical Trade Table](#) (December 20, 2013) for a 1-year history of our trades.

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December 20, 2013
US Interest Rate Strategist

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(as of November 30, 2013)

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December 20, 2013
US Interest Rate Strategist

Stock Rating Category	Coverage Universe		Investment Banking Clients (IBC)		
	Count	% of Total	Count	% of Total IBC	% of Rating Category
Overweight/Buy	995	34%	313	38%	31%
Equal-weight/Hold	1283	44%	388	47%	30%
Not-Rated/Hold	109	4%	26	3%	24%
Underweight/Sell	537	18%	99	12%	18%
Total	2,924		826		

Data include common stock and ADRs currently assigned ratings. An investor's decision to buy or sell a stock should depend on individual circumstances (such as the investor's existing holdings) and other considerations. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months.

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Overweight (O). The stock's total return is expected to exceed the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Equal-weight (E). The stock's total return is expected to be in line with the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
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Underweight (U). The stock's total return is expected to be below the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

Analyst Industry Views

Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.
In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.
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